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COM 111 — Past Questions & Materials

by Renance

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Level ND1 • First Semester • Handout

Week 1 & 2: History, Classification & Impact of Computers

What is a Computer?

A computer is an electronic device that processes data and perform tasks according to a set of instructions

called a program to produce meaningful information. It can store, retrieve, and manipulate data to produce useful information.

Computers can perform a variety of tasks, including calculations, word processing, communication, and

data storage. Computers are used in almost every field today, from education and healthcare to entertainment and business.

Basic Elements of a Computer System

A computer system consists of four main elements:

1. Hardware: The physical parts of the computer. The hardware is divided into:

- Input Devices: Devices used to enter data into the computer, e.g. keyboard, mouse, microphone.
- Output Devices: Devices that display or present information from the computer, e.g. monitor, printer, speakers.
- Processing Unit: The Central Processing Unit (CPU), which acts as the brain of the computer, executing instructions.
- Memory (RAM and Storage): Temporary memory (RAM) stores data while the computer is on, while permanent storage (e.g. hard drive, SSD) saves data long-term.

2. Software: The programs and applications that run on the hardware, such as operating systems (Windows, macOS) and apps (Microsoft Word, Call of Duty, Google Chrome). The computer software is divided into two:

- System Software: These are software that are responsible for managing hardware and creating an environment where application software can run. It works at a fundamental level and ensures that the computer functions properly. Examples of system software are: operating system, device drivers, utility software, firmware.
- Application Software: These are software designed for end-users to perform specific tasks, such as word processing, web browsing, entertainment, etc. Examples of application software are: productivity software (Microsoft Word, Excel, PowerPoint), web browser (Google Chrome, Mozilla Firefox, Microsoft Edge), multimedia software (Adobe Photoshop, VLC Media Player), communication software (WhatsApp, Zoom, Twitter), entertainment software (Sporfy, Call of Duty, Netflix).

3. Data: The raw facts and figures that the computer processes to produce meaningful information.

Data can be:

- Text Data: Letters, numbers, and characters typed into a document.
- Numerical Data: Numbers used in calculations, such as financial records.
- Image Data: Pictures, graphics, and scanned documents.
- Audio Data: Sounds and voice recordings.
- Video Data: Moving pictures and animations.

4. Users: The people who interact with the computer to perform tasks.

History of Computers

Primitive humans used the first counting device. They counted with sticks, stones, and bones. As technology and human intellect advanced over time, more computing devices were created. Some of the

popular computing devices starting with the first to recent ones are described below;

Abacus

The abacus, which is thought to have been the earliest computer, is credited with starting the history of

computers. Abacus is thought to have been created by the Chinese some 4,000 years ago.

The wooden rack has metal rods with beads attached to them. In order to do mathematical computations,

the abacus operator moved the beads in accordance with certain principles. Several nations, including

China, Russia, and Japan, continue to use abacuses.

Napier's Bones

This manually driven calculator was created by Merchiston resident John Napier (1550-1617). He multiplied and divided using nine distinct ivory strips or bones that were numbered in this calculator. Thus,

the instrument was dubbed "Napier's Bones." Additionally, it was the first device to work with the decimal point.

Pascaline

Pascaline is sometimes referred to as an adding machine or an arithmetic machine. It was created by the

French mathematician and philosopher Blaise Pascal between 1642 and 1644. It is thought to have been

the first automatic and mechanical calculator.

In order to assist his father, a tax accountant, Pascal created this device. It was limited to adding and

subtracting. It was a wooden box with wheels and a number of gears. A wheel turns the wheel next to it

when it completes one rotation. To read the totals, a number of windows are provided on top of the wheels.

Leibnitz Wheel

It was created in 1673 by Gottfried Wilhelm Leibnitz, a German mathematician and philosopher. To create

this machine, he modified Pascal's idea. Because it used stepped drums in place of gears, this digital

mechanical calculator was known as the "stepped reckoner."

Difference Engine

In the early 1820s, Charles Babbage—dubbed the "Father of Modern Computer"—designed it. It was a

mechanical computer with basic computation capabilities. It was a steam-powered calculator made to

solve numerical tables, such as logarithm tables.

Analytical Engine

Additionally, Charles Babbage created this calculator around 1830. Punch cards were utilized as input for

this mechanical computer. It has the ability to store data in a permanent memory and solve any mathematical problem.

Tabulating Machine

It was created in 1890 by American statistician Herman Hollerith. Punch cards served as the basis for this

mechanical tabulator. In addition to recording or sorting data or information, it could tabulate statistics.

The U.S. Census of 1890 made use of this device. The Hollerith's Tabulating Machine Company, which

Hollerith founded, changed its name to International Business Machine (IBM) in 1924.

Differential Analyzer

In 1930, the United States saw the introduction of the first electronic computer. Vannevar Bush created

the analogue gadget. This device uses vacuum tubes to switch electrical signals so that computations can

be done. It took only a few minutes to complete 25 calculations.

Mark I

The next significant developments in computer history started in 1937 when Howard Aiken aimed to

create a device capable of carrying out complex mathematical computations. IBM and Harvard collaborated to build the Mark I computer in 1944. It was the first digital computer that could be programmed.

Generations of Computers

The history of computer development is often referred to as the different generations of computing

devices. Each generation of computers is characterized by a major technological development that fundamentally changes the way computer operates, resulting in an increasingly smaller, cheaper, more

efficient and reliable devices. It replaced the gears and other mechanical parts used for counting in previous computing machines. The different generations of computing devices are:

1. First Generation Computers (1940 – 1956): The first computers used vacuum tubes for circuitry and magnetic drums for memory. They were often too huge, occupying entire rooms, very expensive to operate, used a great deal of electricity, and generated a lot of heat which was often the cause of malfunctions. First generation computers relied on machine language to perform operations, and they could only solve one problem at a time. Input was based on punch cards and paper tapes, and output was displayed on printouts. Example of first generation computers are UNIVAC (Universal Automatic Computer) and ENIAC (Electronic Numerical Integrator and Computer).

2. Second Generation Computers (1956 – 1963): Transistors replaced vacuum tubes and ushered in the second generation of computers. The transistor was invented in 1947, but did not see widespread computer use until the late 50s. The transistor was far superior to the vacuum tubes, allowing computers to become smaller, faster, cheaper, more energy efficient, and more reliable than their first generation predecessors. Though the transistors still generated a great deal of heat that subjected the computer to damage, it was a vast improvement over the vacuum tubes. Second generation computers still relied on punched cards for input and printouts for output. Second generation computers moved from cryptic binary machine language to symbolic or assembly languages, which allowed programmers to specify instructions in words. High level programming languages were also being developed at this time such as early versions of COBOL and FORTRAN. These were also the first computers that stored their instructions in their memory which moved from a magnetic drum to a magnetic core technology. The first computers of this generation were developed for the atomic energy industry.

3. Third Generation Computers (1964 – 1971): The development of the integrated circuit was the hallmark of the third generation of computers. Transistors were miniaturized and placed on silicon chips called semi-conductors which drastically increased the speed and efficiency of computers. Instead of punched cards and printouts, users interacted with third generation computers through keyboards and monitors, and interfaced with an operating system which allowed the device to run many different applications at one time with a central program that monitored the memory. Computers for the first time became accessible to the public because they were smaller and cheaper than their predecessors.

4. Fourth Generation computers (1971): The microprocessor brought the fourth generation of computers as thousands of integrated circuits were built into a single silicon chip. What in the first generation filled an entire room could now fit in the palm. The Intel 404 chip developed in the 1971 located all the components of the computer from the central processing unit and memory to input-output controls on a single chip. In 1981, IBM introduced its first computer for the home user and in 1984 Apple introduced the Macintosh. Microprocessors also moved into many areas of life as more and more everyday products began to use microprocessors, as these small computers became more powerful, they could be linked together to form networks which eventually led to the development of the internet. Fourth generation computers also saw the development of GUIs (Graphical User Interfaces), the mouse, and handheld devices.

5. Fifth Generation Computers (present): Fifth generation computing devices, based on artificial

intelligence are still in development, though there are some applications such as voice recognition that are being used. The use of parallel processing and superconductors is helping to make artificial intelligence a reality. The goal of this generation computing is to develop devices that respond to natural language input and are capable of learning and self-organization.

Classification of Computers

Computers can be classified based on their size, mechanism, and purpose/use.

Classification of computer

Based on Size Based on Mechanism Based on Purpose/Use

- ✓ Super computer ✓ Analog computer ✓ General purpose
- ✓ Mainframe computer ✓ Digital computer ✓ Special purpose
- ✓ Mini computer ✓ Hybrid computer
- ✓ Micro computer
- Desktop
- Laptop
- Palmtop

Super Computer

- They are very fast and powerful machines.
- Run at the speed of 10 MIPS (million instructions per second).
- Very expensive.
- Not used for general applications.
- Examples: Cray and CDC Cyber.

Mainframe Computer

- They are built for general computing (i.e. can be used for business or engineering).
- Very fast and will process information at about 10 MIPS. They have large storage capacity.
- Very expensive.
- Not easily found in general design offices.
- They can handle 100s – 1000s of users at the same time.

Mini Computer

- Developed in the 1960's resulting from advances in microchip technology.
- Smaller and less expensive than mainframe computers.
- Run at several MIPS and can support 5-20 users.
- Low cost and high performance.
- Used in small organizations.

Micro Computer

- Invented in the 1970's.
- Generally used for home and personal computing.
- Dedicated data processing workstations.
- Handle one user at a time.
- They are cheap.

Desktop computer is a personal computer designed to fit on a small office desk. It includes: monitor, keyboard, mouse, system unit, etc.

Laptop is a personal computer designed for mobility. It is small and light to place on the lap of the user. It integrates most of the components of the desktop.

Palmtop is a small computer that literally fits on your palm. Due to their small size, they are limited in capabilities.

Analog Computer

- The word <analog= means continuously changeable in quantity.
- The analog computers accept input data in continuous form and output is obtained in the form of graphs. Example: voltage, current, sound, speed, temperature, pressure, etc.
- The analog computers are used to measure the continuous values. Example: thermometer, speedometer, voltmeter, clock, etc.

Digital Computer

- The word <digital= means separate. It refers to binary system, which consists of only two discrete digits, i.e. 0 and 1.
- Digital data consists of binary data represented by OFF (low) and ON (high) electrical pulse.
- In digital computers, quantities are counted rather than measured.
- A digital computer operates by counting numbers or digits and give output in digital form. Examples: calculators, personal computers, digital watches, digital thermometer.
- They are used in offices, education, hotels, hospitals, data processing, finance, etc.

Hybrid Computer

- They combine the feature of both analog and digital computers.
- In hybrid computers, the user can process both the continuous (analog) and discrete (digital) data by converting from one form to another using analog-to-digital converter (ADC) and digital-to-analog converter (DAC).
- These are special purpose computers, such as a machine that measures patients heart beat and convert the analog data into digital form to display on the screen, or dispenser machines at petrol pumps that converts the measurements of fuel flow into quantity and price.
- They are very fast and accurate.
- These are used in scientific fields such as weather forecasting, airline systems, etc.

Special Purpose Computer

- These are designed to perform a specific task. Such computers lack flexibility.
- They perform the task for which they are designed very efficiently.
- Examples: air craft control system, missile guidance system, heart beat monitor, etc.

General Purpose Computer

- A general purpose computer is one that can be used for a variety of applications.
- Its creativity enables execution of programs of almost any type. These are used in business

applications such as word processing, spreadsheet, database, games, communications, etc.

Benefits of Computers to the Society

Computers have become an integral part of modern society, transforming many aspects of our lives with

their versatility and efficiency.

The benefits of computers span many different sectors, including education, business, healthcare, and

personal development. Understanding these benefits highlights the integral role computers play in shaping an advanced and connected world.

1. Education

- E-learning platforms enable online learning.
- Digital libraries provide access to research materials.
- Examples: Coursera, Udemy, YouTube, Khan Academy

2. Communication

- Emails, instant messaging, and video calls connect people worldwide.
- Examples: WhatsApp, Zoom, Gmail.

3. Healthcare

- Computers help in diagnosing diseases, managing patient records, and conducting surgeries.
- Examples: MRI Scanners, Electronic Medical Records (EMR).

4. Business and Finance

- Automate tasks such as payroll, accounting, and stock trading.
- Examples: Banking software, E-commerce platforms like Amazon, Jumia.

5. Transportation

- GPS systems help navigation.
- Airline and railway booking systems use computers for scheduling.
- Examples: Google Maps, Air Traffic Control Systems.

6. Entertainment

- Computers are used in gaming, movie production, and music streaming.
- Examples: Netflix, Spotify, PlayStation.

Social Implications of Computers on Society

While computers bring many benefits, they also have significant social implications, particularly in privacy

and quality of life.

Privacy Concerns

- Personal data is collected by various websites, social media, and apps, raising concerns about data security.
- Cybercrime (hacking, phishing, identity theft) has increased.
- Governments and companies monitor online activities (e.g. surveillance systems).

Impact on Quality of Life

- Social isolation due to excessive screen time.
- Online communication have changed how people interact, with virtual relationships replacing real-life interactions, leading to social isolation.
- Health issues like eye strain, obesity, and poor posture due to prolonged computer use.
- Job losses due to automation replacing human labor.

Digital Divide

- Not everyone has equal access to computers and the internet, leading to disparities in education, employment, and communication.
- Wealthier individuals and developed countries benefit more from technology than poorer regions.

Ethical Issues

- The rise of AI and automation raises ethical concerns about surveillance, deepfakes, and bias in decision-making systems.
- Misuse of technology for illegal activities (e.g. cyberbullying, spreading fake news).

Environmental Impact

- The production and disposal of computers contribute to electronic waste (e-waste).
- Data centers consume large amounts of energy, increasing carbon footprints.

Week 3: Computer Hardware

Introduction to Computer Hardware

Computer hardware refers to the physical components of a computer system that you can see, touch, and

interact with. These components work together to perform tasks, process data, and display output.

Without hardware, a computer cannot function.

Types of Computer Hardware and their Functions

Hardware components can be classified into different types based on their roles in the computer system:

Input Devices

Input devices allow users to enter data and instructions into the computer. They serve as a bridge between

humans and computers.

- Keyboard: Used for typing text, numbers, and special characters.
- Mouse: A pointing device used to navigate and interact with objects on the screen.
- Scanner: Converts physical documents or images into digital form.
- Microphone: Captures sound and converts it into digital data.
- Webcam: Captures images and videos for video conferencing or recording.
- Joystick/Game Controller: Used for gaming and controlling movement in virtual environment.

Output Devices

Output devices allow the computer to display, print, or produce information to the user in the desired

format.

- **Monitor (Screen/Display):** Displays visual information from the computer.
- **Printer:** Produces a hard copy of digital documents.
- **Speaker:** Converts digital audio signals into sound.
- **Headphones:** Output sound privately to a single user.
- **Projector:** Displays images or videos on a large screen.

Processing Unit (Central Processing Unit – CPU)

The CPU (Central Processing Unit) is the <brain> of the computer. It processes all instructions and performs

calculations necessary for running applications. The CPU consists of three main components:

- **Control Unit (CU):** Directs the flow of data and instructions within the computer.
- **Arithmetic and Logic Unit (ALU):** Performs mathematical calculations and logical operations.
- **Registers:** Temporary storage areas within the CPU that hold data being processed.

The speed of the CPU is measured in gigahertz (GHz), and a higher GHz value means faster processing power.

Storage Devices

Storage devices are used to store data, programs, and files either temporarily or permanently. They are

categorized into four:

- **Random Access Memory (RAM):** It is the primary memory of the computer that stores data temporarily. It stores data that is currently in use. It is volatile (data is lost when power is off).

Example of such memory are DDR 3, DDR 4, DDR 5 RAM.

- **Read-Only Memory (ROM):** It is a permanent memory that stores essential system instructions. It stores firmware and boot-up instructions (BIOS). It is non-volatile (data remains even when power is off). It is fast but slower than RAM. Examples of such memory are BIOS, firmware chips. You cannot modify the data in ROM, but can only read from it.

- **Secondary Storage:** It is a permanent storage for user data, files, and software. It stores data long-

term, even when the computer is off. It is non-volatile, and is slower than RAM and ROM. Example of such storage are HDD, SSD, USB flash drives.

- **Auxiliary Memory:** Another term for secondary storage; sometimes includes external storage. Provides extra storage for data, backups, and large files. It is non-volatile, and is slower than RAM and ROM. Example of such storage are HDD, SSD, USB flash drives.

Other important hardware components include:

- **Motherboard:** The motherboard is the main circuit board that connects all computer components, including the CPU, RAM, storage, and input/output devices.
- **Power Supply Unit (PSU):** The power supply unit converts electricity from an external source (e.g. NEPA, BEDC, EEDC) into a form that the computer can use.
- **Cooling Systems (Fans & Heat Sink):** Computers generate heat while operating. Cooling Systems, such as fans and heat sinks, prevent overheating by dissipating heat.

Week 4: Central Processing Unit

What is a Central Processing Unit (CPU)?

The Central Processing Unit (CPU), often referred to as a computer processor, microprocessor or simply

processor, is the brain of the computer. It provides processing power to the computer system. It carries

out instructions from software programs by performing basic arithmetic, logic, control, and input/output

operations. Without the CPU, a computer cannot function.

It performs most of the processing operations inside a computer, executing instructions and managing

data manipulation. The CPU fetches and execute instructions from the primary memory (RAM), coordinating the activities of all the hardware components to perform various tasks.

There are two main computer processor manufacturers: Intel and AMD (Advanced Micro Devices). These

two companies produce most of the processors used in desktop, laptops, and notebooks.

A CPU is placed on the mother motherboard in the CPU socket with a locking lever mechanism in order to

fix the CPU properly into the CPU socket. A heat-sink is mounted on top of the CPU chip which protects

the CPU from the excessive heat generated by the CPU. The heat-sink absorbs the heat, and is then cooled

by a fan placed on top of the heat-sink.

Main Components of the CPU

The performance of a CPU is influenced by many factors like architecture, clock speed, cache size, and the

number of cores. The arithmetic logic unit, control unit, and registers are the core parts of the CPU.

Others

are referred to as auxiliary units as they enhance the CPU efficiency by managing memory, speeding up

processing, and handling complex operations.

1. Control Unit: The control unit acts as the manager of the CPU. It directs the flow of data between the CPU and other parts of the computer, telling the ALU and memory how to respond to the instructions it receives. It fetches, decodes, and execute instructions from memory.

2. Arithmetic Logic Unit (ALU): The ALU performs all the mathematical calculations and logical operations. Arithmetic operations include basic calculations like addition, subtraction, multiplication, and division. Logical operations involve comparing values (like determining if one number is greater than or equal to another).

3. Registers: Registers are small, fast storage locations within the CPU that hold data temporarily while the CPU processes it. There are different types of registers, such as:

a. Accumulator (AC): Stores intermediate results during computations.

b. Program Counter (PC): Keeps track of the next instruction to be executed.

- c. Instruction Register (IR): Holds the current instruction being executed.
- d. Memory Address Register (MAR): Holds the address in memory where data is to be read from or written to.
- 4. Cache: The cache is a small amount of high-speed memory that stores frequently accessed data and instructions to speed up processing. It is closer to the CPU than regular RAM, so it allows for quicker access to critical data. There are three levels of cache:
 - a. L1 Cache: The smallest and fastest, directly integrated into the CPU core.
 - b. L2 Cache: Larger but slower than L1, still within the CPU chip.
 - c. L3 Cache: Larger and shared among all CPU cores, slower but larger than L2.
- 5. Clock: The clock in the CPU generates the electrical pulses that synchronize the operations of the CPU. The clock speed is measured in GHz (gigahertz), and it determines how many instructions the CPU can process per second. A faster clock speed generally means a faster CPU.
- 6. Bus Interface: This is responsible for managing the flow of data between the CPU and other components, such as memory or input/output devices. There are different types of buses:
 - a. Data bus: Transfers data between the CPU and other components.
 - b. Address bus: Carries the addresses of where data is to be fetched or sent.
 - c. Control bus: Carries control signals from the control unit to other parts of the computer.
- 7. Instruction Decoder: The instruction decoder interprets the instructions fetched from the memory and converts them into signals that the ALU and other components can understand and act upon.
- 8. Floating Point Unit (FPU): The FPU handles complex arithmetic involving floating-point numbers (decimals). It's separate from the ALU, which primarily handles integer operations.
- 9. Cores: Modern CPUs have multiple cores, which are individual processing units within the CPU. A multi-core CPU can handle multiple tasks simultaneously (parallel processing), significantly improving performance for multitasking or heavy workloads.

How the CPU Works

The CPU operates through a cycle called the fetch-decode-execute cycle.

- Fetch: Retrieves an instruction from the memory.
- Decode: Interprets what the instruction means.
- Execute: Carries out the instruction and sends results to the appropriate part of the computer.

Functions of the CPU

- To perform arithmetic operations.
- To perform logical operations.
- To control functions of other hardware components.
- To fetch the data and program instructions from main memory RAM.
- To decode the program instructions.
- To operate on the data as per program instructions.
- To store the processed data into the RAM.
- To continuously execute instructions.

Types of Computer Memory

- Random Access Memory (RAM): It is the primary memory of the computer that stores data

temporarily. It stores data that is currently in use. It is volatile (data is lost when power is off).

Example of such memory are DDR 3, DDR 4, DDR 5 RAM.

- **Read-Only Memory (ROM):** It is a permanent memory that stores essential system instructions. It stores firmware and boot-up instructions (BIOS). It is non-volatile (data remains even when power is off). It is fast but slower than RAM. Examples of such memory are BIOS, firmware chips. You cannot modify the data in ROM, but can only read from it.

- **Secondary Storage:** It is a permanent storage for user data, files, and software. It stores data long-

term, even when the computer is off. It is non-volatile, and is slower than RAM and ROM. Example of such storage are HDD, SSD, USB flash drives.

- **Auxiliary Memory:** Another term for secondary storage; sometimes includes external storage. Provides extra storage for data, backups, and large files. It is non-volatile, and is slower than RAM and ROM. Example of such storage are HDD, SSD, USB flash drives.

Functions of Auxiliary Memory

Auxiliary memory (also called secondary storage) refers to devices that store data permanently, even

when the computer is turned off. Its functions include:

- **Long-term data storage:** Saves files, applications, and operating systems.
- **Backup and recovery:** Protects data from loss due to system failure.
- **Data transfer:** Moves files between devices using external storage (e.g. USB drives).
- **Virtual memory support:** Acts as extra storage when RAM is full, helping the system run smoothly.

Measurements of Storage

Computer storage is measured in bytes, with larger units increasing in powers of 1,024.

1. **Bit (b):** The smallest unit of data representing either 0 or 1.
2. **Byte (B):** 8 bits make up one byte, enough to store one character (e.g. <A=, <1=, <@=).
3. **Kilobyte (KB):** 1,024 bytes (approximately 1,000 bytes).
4. **Megabyte (MB):** 1,024 KB (approximately 1 million bytes).
5. **Gigabyte (GB):** 1,024 MB (approximately 1 billion bytes).
6. **Terabyte (TB):** 1,024 GB (approximately 1 trillion bytes).
7. **Petabyte (PB):** 1,024 TB (approximately 1 quadrillion bytes).
8. **Exabyte (EB), Zettabyte (ZB), Yottabyte (YB):** Extremely large storage capacities used in data centers and cloud storage.

Week 5: Concept of Computer Software

What is Computer Software?

Computer software is basically a program that tells the computer's hardware what to do.

First let's understand that hardware devices need user instruction to function. A set of instructions that

achieve a single outcome are called a program or procedure.

So, a computer software can be defined as a collection of programs, data, and instructions that tell a

computer how to perform specific tasks. Unlike hardware, which refers to the physical components of a

computer, software consists of code and digital instructions that enable hardware to function.

Types of Software

Computer software is broadly categorized into two main types: system software and application software.

User

Application Software

System Software

Computer Hardware

System Software

System software is a software that controls a computer's internal functioning, through an operating

system, and also controls such peripherals such as monitors, printers, and storage devices. It is also responsible for creating an environment where application software can run. It works at a fundamental

level and ensures that the computer functions properly.

Characteristics of system software include:

- Runs in the background and starts when the computer is powered on.
- Directly interacts with the hardware components (CPU, memory, storage, etc).
- Provides a platform for users to install and run application software.
- Typically developed using low-level programming languages like C and Assembly.

Types of System Software

1. Operating System (OS): This is the most important system software that manages all computer resources. It is usually the first software to be loaded into the computer memory. It acts as an intermediary between the user of a computer, application software and the computer hardware. It manages the computer's basic functions like storing data in memory, retrieving files from storage devices, scheduling tasks based on priority, etc. Examples: Windows, MacOS, Linux, Android & iOS (mobile).

2. Firmware: This is a low-level software that's embedded in hardware like ROM chips to control basic functions. Examples are:

- a. BIOS (Basic Input/Output System): This manages the startup settings before the OS loads.
- b. UEFI (Unified Extensible Firmware Interface): The modern replacement for BIOS. Provides better encryption and GUI to navigate.

3. Device Driver: This is a special software that enables the operating system to communicate with hardware devices. It controls and monitors functioning of a specific device on the computer. Each device attached to the computer has a specific driver associated with it. When you attach a new device, you need to install its driver so that the OS knows how to manage the device. Examples: printer driver, graphics driver, sound driver.

4. Utility Software: Helps maintain the system and improve performance. Examples: antivirus software, disk management tools, file management tools, compression tools, backup tools.

Application Software

Application software is a software designed for end-users to perform specific tasks, such as word processing, web browsing, entertainment.

Characteristics of application software include:

- Runs on top of system software.
- Designed for a specific function like editing documents, playing games, browsing the internet.
- Can be installed, updated, and removed as needed.
- Developed using high-level programming languages like Python, Java, C++, etc.

Types of Application Software:

1. **Productivity Software:** Used for office and business tasks. Examples: Microsoft Suite (Word, Excel, PowerPoint, Access), Google Workspace (Gmail, Docs, Sheets, Slides).
2. **Web Browsers:** Allows the user to access and navigate the internet. Examples: Google Chrome, Mozilla Firefox, Microsoft Edge.
3. **Multimedia Software:** Used for creating, editing, and playing media. Examples: Adobe Photoshop, VLC media Player.
4. **Communication Software:** Enables users to communicate via text, voice, video. Examples: WhatsApp, Zoom, Skype.
5. **Entertainment and Gaming Software:** Used for leisure and gaming purposes. Examples: Spotify, Netflix, Call of Duty.
6. **Educational Software:** Helps users learn new skills or subjects. Examples: Duolingo, Khan Academy, MATLAB.

Key differences between system software and application software.

Feature	System Software	Application Software
Purpose	Manages the hardware and provides a platform for applications	Helps users perform specific tasks
User Interaction	Runs in the background; users rarely interact with it directly	Users interact with it directly
Dependency	Needed for the computer to function	Runs on system software
Examples	Operating Systems, Drivers, Utility	Microsoft Word, Chrome, Photoshop
Execution	Starts when the computer boots and runs continuously	Starts when opened and closes when exited

Computer Languages

Computer language is the language used to communicate with the computer.

Computer language is commonly referred to as programming language. A programming language is a set

of rules, symbols, and syntax used to write instructions that a computer can understand and execute.

People who write in this language are called programmers. This allows programmers to create software,

applications and systems by providing the commands the computer should follow.

Computer language is divided into two: low-level language and high-level language.

Low-Level Languages

These languages interact directly with the hardware and are faster but harder to understand. There are

two types of low-level programming languages:

- **Machine Language:** Computers can understand only digital signals, which are in binary digits (0 and 1). So, the instruction given to the computer can be only in binary codes. This is the only language the computer directly understands. Writing a program in machine level language is a difficult task because it is not easy for computer to write instructions in binary codes. There is no need for any translator, since the computer is already communicating in the machine level language. Example: 01000001 01000010.

- **Assembly Language:** The difficulties faced in machine level language were reduced to some extent

by using a modified form of machine level language called assembly language. In assembly language, instructions are given in English like words (mnemonics) such as MOV, ADD, SUBB, etc instead of raw binary. Assembly language is hardware-dependent, meaning it is not easily portable across different types of computers. Assembly language is translated into machine level language using a translator known as <assembler>. Example: MOV A, B (moves data from register B to A).

Assembly Machine

Assembler

Language Language

High-Level Language

High-level language is a programming language that is written in human readable language. It is easier

to read, write, and maintain. It is portable, and not designed to run on just one type of machine. A high-

level language is translated into a machine language using a compiler or an interpreter. Example of high-

level language are Python, Java, C++, JavaScript.

High-Level Machine

Compiler/Interpreter

Language Language

Source and Object Programs

In programming, when we write code, it needs to be converted into a form that the computer can understand. This process involves two key terms: source program and object program.

Source Program

A source program is the original code written by a programmer in a high-level language or assembly

language. This code is written in a human-readable format and must be converted into machine language

before execution. This program can be modified and debugged easily.

Object Program

An object program is the translated version of a source program, written in machine code (binary), which

the computer can execute. It is generated by an assembler, compiler, or interpreter from the source code.

The object program is not human readable, but is understandable by the computer, hence it can be executed directly by the computer. It can be linked with other object files to form an executable file.

So, we can say the source program is what the programmer writes. And the object program is the machine-code version of the source program, which the computer can execute.

Language Translators

A translator is a program or processor that converts code written in one programming language into another language, typically from a high-level or low-level language into machine code that the computer

can execute.

There are three main types of translators:

- Assembler
- Interpreter
- Compiler

Each type of translator works differently and is used for specific types of programming languages.

Assembler

Assembler is a translator program that is used to convert an assembly language into a machine language.

The assembler converts the assembly language mnemonics into machine code that the CPU can execute.

They are fast in translation since it works with low-level instructions. The output is hardware-specific,

meaning different processors require different assembly languages.

Interpreter

Interpreter is a translator program that is used to convert high-level language program into a machine

language. It translates the program line by line rather than all the code at once. If there is an error in any

line, the execution stops at the line. It has slower execution time because translation happens during

runtime. Also, it produces no separate executable file; the source code must be interpreted every time it

runs.

Compiler

Compiler is a translator program that is used to convert high-level language program into a machine

language. It translates the whole program at once and generates the object code or file with a list of errors

if present. The object file is then linked with libraries to produce an executable file (.exe). The computer

can run the executable file without needing the original source code.

Computer Packages and user Application Software

Both computer packages and user application software refer to programs designed to perform specific

tasks on a computer. However, they serve different purpose and are used in different contexts.

Computer Packages

A computer package is a set of pre-designed software programs that work together to perform specific

functions. These packages are usually bundled and provide a complete solution for users in various industries. Examples:

Category Examples Purpose

Office Suite Microsoft Office, Google workspace Word processing, spreadsheet, presentations

Graphics and Design Adobe creative suite, CorelDraw Photo editing, graphic design, video editing

Database Microsoft access, Oracle, MySQL Storing and managing large data

Management

Statistical and SPSS, MATLAB, R Data analysis, research, and

Scientific engineering simulations.

User Application Software

User application software refers to individual programs designed to help users perform specific tasks on

a computer. These can be standalone or part of a larger software package.

Category Examples Purpose

Word processing Microsoft Word, Google docs Creating and editing documents

Spreadsheet Microsoft Excel, Google Sheets Data analysis and financial calculation

Web browser Google Chrome, Mozilla Firefox Accessing the internet

Week 6: Data Processing

What is Data Processing?

Data in its raw form is not useful to any organization. Data processing is the method of collecting raw data

and translating it into usable information. It involves collecting, organizing, analyzing, storing, and presenting data in a way that is useful for decision-making.

In simpler terms, it is the method by which raw facts (data) are converted into meaningful outputs

(information) through computational or manual operations.

Data Processing Cycle

The data processing cycle consists of a series of steps where raw data (input) is fed into a system to produce actionable insights (output). Each step is taken in a specific order, but the entire process is repeated in a cyclic manner. The first data processing cycle's output can be stored and fed as the input for the next cycle, as the illustration below shows us.

Fig: Data processing cycle

Generally, there are six main steps in the data processing cycle:

Step 1: Collection

The collection of raw data is the first step of the data processing cycle. The type of raw data collected has a huge impact on the output produced. Hence, raw data should be gathered from defined and accurate sources so that the subsequent findings are valid and usable. Raw data can include monetary figures, website cookies, profit/loss statements of a company, user behavior, etc.

Step 2: Preparation

Data preparation or data cleaning is the process of sorting and filtering the raw data to remove unnecessary and inaccurate data. Raw data is checked for errors, duplication, miscalculations or missing data, and transformed into a suitable form for further analysis and processing. This is done to ensure that only the highest quality data is fed into the processing unit.

The purpose of this step is to remove bad data (redundant, incomplete, or incorrect data) so as to begin assembling high-quality information so that it can be used in the best possible way for business intelligence.

Step 3: Input

In this step, the raw data is converted into machine readable form and fed into the processing unit. This can be in the form of data entry through a keyboard, scanner or any other input source.

Step 4: Data Processing

In this step, the raw data is subjected to various data processing methods using machine learning and artificial intelligence algorithms to generate a desirable output. This step may vary slightly from process to process depending on the source of data being processed (data lakes, online databases, connected devices, etc.) and the intended use of the output.

Step 5: Output

The data is typically transmitted and displayed to the user in a readable form like graphs, tables, vector files, audio, video, documents, etc. This output can be stored and further processed in the next data processing cycle.

Step 6: Storage

The last step of the data processing cycle is storage, where data and metadata are stored for further use.

This allows for quick access and retrieval of information whenever needed, and also allows it to be used as input in the next data processing cycle directly.

Types of Data Processing

Data processing techniques refer to the different ways data can be handled or processed based on the

mode of operation. They define how data flows and is transformed. It focuses on when and how data is

structured and scheduled for processing – whether all at once (batch), instantly (real-time), across multiple

computers (distributed), or shared among users (time-sharing).

Batch Processing

In batch processing, data is collected over a period and processed at a later time as a single unit. It is used

for tasks that do not require immediate updates.

- It is cost effective for large data sets
- Can be scheduled (e.g. nightly, weekly processing)
- Suitable for repetitive tasks

Examples:

- Payroll Processing: Employee salaries are calculated and paid at the end of the month.
- Bank Transaction Processing: Banks process multiple transactions in batches at night.
- Sales Report Processing: A supermarket collects sales data throughout the day and processes all transactions at night.

Real-Time Processing

In real-time processing, data is processed immediately as it is received, without delay. It is commonly used

in applications where quick decision-making is required.

- It is extremely fast
- Processes data as events occur
- Requires high computing power

Examples:

- ATMs: Updates account balances immediately after a withdrawal.

- Stock Market Trading: Processes transactions in milliseconds.
- Online Gaming: Real-time game data updates for players worldwide.
- Inventory Management System: The cash register updates inventory immediately customer make purchases.

Online Processing

In online processing, data is stored in a database and is processed over a network, often in real time or later in batches.

- High speed and accuracy
- Can handle large datasets
- Uses modern computing technologies like databases, cloud computing, and artificial intelligence

Examples:

- Online Banking: Customers account balances are updated in real time as you transfer money.
- Online Stores: An online grocery store updates stock levels as customers place orders.

Time-Sharing Processing

In time-sharing processing, multiple users or processes share computing resources simultaneously. The

system allocates small time slots (CPU time slices) to each task in a rotating manner. This method allows

multiple users to interact with a computer system simultaneously without significant delays.

- Many users can access and use the system at the same time
- The processor switches between tasks rapidly, ensuring fair usage
- The system responds almost instantly to user inputs
- Maximizes the use of CPU and memory

Examples:

- Online Banking Systems: Multiple users log into the bank's website at the same time. The server assigns processing time to each user request, ensuring smooth transactions.
- Airline Reservation Systems: Hundreds of users book flights online. The system processes each request in time slices, preventing long wait times.
- Cloud Computing Services: Many users access and edit documents stored in the cloud at the same time. The server distributes processing power to handle multiple requests efficiently.

Distributed Processing

In distributed processing, multiple computers work together to process data across different locations. It

is commonly used in cloud computing and large organizations.

- Increases efficiency by sharing workloads
- Reduces processing time
- Enables remote access to data

Examples:

- Cloud Storage Services (Google Drive, OneDrive): Multiple servers handle data storage and

processing.

- Social Media Platforms: Facebook, Instagram, X (Twitter) processes millions of user posts across global servers.
- Sales Data: A supermarket chain processes transactions from all its branches using a central database.

Parallel Data Processing

In parallel processing, large task is divided into smaller sub-tasks that run simultaneously on multiple

processors to speed up processing time.

- Significantly reduces processing time
- Used in high-performance computing
- Requires specialized hardware

Examples:

- Weather Forecasting: Supercomputers analyze weather patterns simultaneously.
- Scientific Simulations: Used in space research and artificial intelligence training.

Data Processing Methods

Data processing methods refer to the techniques or tools used to process data, whether manually, machines, or fully automated.

There are three main data processing methods: manual, mechanical, or electronic.

Manual Processing

This method involves human handling and processing data without machines. It includes writing, filling,

and calculating manually. Example: A small shopkeeper writing sales records in a notebook.

Advantages

- Low Cost: No need for expensive machines or software.
- Easy to Use: No technical skills or training required.
- Flexibility: Can be used in places without electricity or computers.
- Security: Less risk of hacking or cyber threats.

Disadvantages

- Slow and Time-Consuming: Processing large amounts of data takes a long time.
- Prone to Errors: High chance of mistakes due to human input.
- Difficult to Retrieve Data: Finding old records manually is tedious.
- Limited Scalability: Cannot handle large data efficiently.

Mechanical Data Processing

This method uses simple machines like typewriters, calculators, and cash registers to assist in data

processing. Example: A cashier using a mechanical cash register to total up bills.

Advantages

- Faster than Manual Processing: Machines speed up calculations and data entry.
- Reduces Human Error: More accuracy in calculations.

- Easier Record Keeping: Printed documents last longer than handwritten ones.

Disadvantages

- Limited Automation: Still requires a lot of human effort.
- Maintenance Cost: Machines need servicing or replacement.
- Storage Issues: Physical records take up space.

Electronic Data Processing

This method uses computers and software to process data automatically, making it the most efficient method today. Example: A supermarket using barcode scanner and a computer-based inventory system.

Advantages

- High Speed: Computers process data in milliseconds.
- High Accuracy: Reduces human error significantly.
- Large Data Handling: Can process massive amounts of data.
- Easy Data Retrieval: Stored data can be accessed instantly.
- Automation: Reduces manual workload.
- Cost-Effective in the Long Run: It saves time and money.

Disadvantages

- High Initial Cost: Expensive to set up hardware and software.
- Technical Knowledge Required: Users need training to operate computers.
- Security Risks: Data can be hacked, lost, or corrupted.
- Dependence on Power and Internet: System fail without electricity or network access.

Week 7: Computer Operations

Operating a computer system involves a number of procedures that makes sure the system runs effectively and securely. This entails handling files, correctly turning the system on and off, and carrying out necessary maintenance like formatting storage media. These operations aid in preventing hardware damage, data loss, and system breakdowns.

Some operations that can be carried out on a computer includes:

Starting Up (Booting the Computer)

Booting is the process of turning on a computer and loading the operating system (OS). It can be categorized into two types:

- Cold Booting: Starting a computer that is completely turned off.
- Warm Booting: Restarting a computer that is already on.

Steps to Start a Computer Properly

1. Check Power Supply: Make sure the computer is connected to a power source. If using a laptop, make sure the battery is charged.
2. Turn On the Computer: Press the power button on the system unit or the laptop.
3. POST (Power-On Self-Test): The system checks its hardware components (e.g. RAM, hard drive,

etc) for issues.

4. Load the Operating System: The operating system starts loading, displaying a welcome screen.

5. Login (If Required): Enter a password (if set) to access the password.

Shutting Down the Computer

Turning off the computer properly is important to prevent data loss and system corruption.

Steps to Shut Down the Computer Properly

1. Close All Applications: Save your work and close all open programs.
2. Access the Shutdown Menu: Click the Start Menu, select Power, then click Shut Down.
3. Wait for Shutdown Process: The system will close processes and turn off automatically.
4. Turn Off External Devices: If you're using a desktop, turn off other peripherals like monitor, printer, speaker, and unplug if necessary.

Restarting the Computer

Restarting (or rebooting) a computer is necessary when installing updates, troubleshooting problems, or refreshing the system.

Steps to Restart a Computer

1. Click Start Menu → Power → Restart.
2. The computer will shut down and turn on automatically.

Formatting a Storage Device (Hard Drive, USB, SD Card)

Formatting a storage device prepares the storage device for use by erasing all data and setting up a file

system. It is necessary when installing a new operating system or fixing a corrupted drive.

Steps to Format a USB Drive

1. Insert the USB drive into the computer.
2. Open <This PC> or <My Computer> and locate the drive.
3. Right click on the drive and select Format.
4. Select the File System: NTFS (for Windows), FAT32 (compatible with other OS).
5. Click <Start> and proceed.
6. Wait for the process to complete.

NB: Formatting erases all the data in the storage device, so make sure you back up important files before formatting.

Basic File Operations

Beyond startup and shutdown, users need to understand how to carry out basic file operations.

a. Creating and Saving File

- Open a program (e.g. Microsoft Word).
- Type your content.
- Click File → Save and choose a location (or use Ctrl + S).
- Give the file a name and select a format (e.g. .docx, .pdf).

b. Copying and Deleting Files

- Copy: Right click on a file. Select Copy. Go to destination folder, right click and paste.
- Delete: Right click on a file Select Delete. The file moves to the Recycle Bin.

Installing and Uninstalling Software

a. Installing Software

- Download the setup file (.exe file).
- Double click the file to open the installer.
- Follow the installation instructions.
- Click Finish after installation.

b. Uninstalling Software

- Go to Control Panel → Programs and Features.
- Locate the software.
- Click Uninstall and follow prompts.

Week 8 & 9: Computer Security

Computer security, also called cybersecurity, is the protection of computer systems, networks, and information from harm, theft, unauthorized use, damage, and disruption. It involves implementing security measures to safeguard hardware, software, and information from cyber threats such as viruses, hacking, phishing, and malware attacks.

Computer hardware is typically protected by the same means used to protect other valuable or sensitive

equipment—namely, serial numbers, doors and locks, and alarms.

Objectives of Computer Security

1. Confidentiality: Ensuring that sensitive information is accessible only to authorized users.
2. Integrity: Maintaining the accuracy and reliability of data, preventing unauthorized modification.
3. Authentication: Verifying the identity of users before granting access.
4. Availability: Ensuring that computer systems and data remain accessible when needed.

Key Aspects of Computer Security

1. Access Control: This involves managing who has access to what resources within a system or network. Access control mechanisms include passwords, biometric authentication, access control lists (ACLs), and role-based access control (RBAC).
2. Encryption: Encryption is the process of converting data into a format that cannot be easily read or understood by unauthorized users. It helps protect data confidentiality by scrambling information so that only authorized parties can access it.
3. Firewalls: Firewalls are security devices or software that monitor and control incoming and outgoing network traffic based on predetermined security rules. They serve as a barrier between a trusted internal network and untrusted external networks, such as the internet, filtering the traffic to prevent unauthorized access.
4. Antivirus and Anti-Malware Software: These programs are designed to detect, prevent, and remove malicious software (malware) such as viruses, worms, Trojans, and ransomware from computer systems.
5. Intrusion Detection and Prevention Systems (IDPS): IDPS monitor network or system activities

for malicious activities or policy violations and can take actions to prevent or mitigate potential threats.

6. Patch Management: Regularly updating software and firmware with patches provided by vendors

helps fix known vulnerabilities and protect systems from exploitation by attackers.

7. Security Policies and Procedures: Establishing and enforcing security policies and procedures help

ensure that employees and users follow best practices for maintaining security, such as password management, data handling, and incident response protocols.

8. Incident Response Plan (IRP): A set of procedures to detect, respond to, and recover from security

breaches.

9. Disaster Recovery Plan (DRP): A set of plan to ensure business continuity in case of cyberattacks, hardware failures, or natural disasters.

Basic Safety for Computer Room

The safety of a computer room is the safety of computer systems and other things found in a computer room.

1. A computer room must be spacious.
2. A computer room must be well ventilated.
3. A computer room must be clean and well lit.
4. A computer room temperature must be cool.
5. No one must eat or drink inside the computer room, not even the teacher.
6. Each computer system must have their own UPS.
7. Backup power must be available.
8. Fire extinguisher must be available in a computer room.
9. Safety rules must be visible for all users.
10. Turn off the power and unplug equipment before performing service.

Tips to Minimize Risk of Computer-Related Accident:

- i. Site your computer near an electric socket to avoid trailing wires across the floor; if you use an extension cable, make sure it doesn't overheat and nobody can trip over it.
- ii. Take care not to overload electric sockets; use trailing multi-socket units rather than plug adapters.
- iii. Always follow installation and service instructions in your computer guidebook closely. If in doubt, leave it to the experts.
- iv. Electricity and water do not mix - keep drinks and plants well away from computers.
- v. Regularly check all electrical equipment for damaged plugs or frayed cables.
- vi. Computers are large and bulky pieces of equipment, move them only if you feel confident in doing so, and with care, especially up and down stairs. Use a trolley and a lifter and ask for help.
- vii. Do not allow children to move computers.
- viii. Do not allow children to play on or with computer swivel chairs.

ix. Make sure the computer is sited in a position where you have plenty of room to move and to get out of the room in an emergency.

Malware Infection and Prevention

Malware, short for malicious software, is a broad category of software designed to harm, exploit, or

otherwise compromise the data and systems of its victims. Common types of malware include viruses,

worms, trojans, ransomware, spyware, adware, and rootkits. Each type of malware operates differently

but often has overlapping features and objectives, such as stealing information, disrupting operations, or

extorting money.

Common types of malwares include:

1. Viruses: Malicious code that attaches itself to legitimate programs and replicates when the host program runs.
2. Worms: Self-replicating malware that spreads through networks without user intervention.
3. Trojans: Malicious programs disguised as legitimate software that trick users into executing them.
4. Ransomware: Encrypts files and demands payment for the decryption key.
5. Spyware: Secretly monitors and collects user activity and data.
6. Adware: Automatically displays or downloads advertising material.
7. Rootkits: Designed to gain unauthorized root or administrative access to a system.

Infection Vectors

Infection vectors are the different ways malware or cyber threats enter a system, network, or device.

Hackers and cybercriminals use these methods to gain access, steal data, or cause damage.

Common infection vectors include:

1. Phishing Emails & Malicious Links: An email that looks legitimate but contains a link or attachment that, once clicked, installs malware.
2. Malicious Attachments & File Downloads: Downloading a file from an untrusted source can introduce malware into your system.
3. Drive-By Download: Some websites are rigged to install malware the moment you visit them – no clicking is required.
4. Removable Storage Media (USBs, External Drives, CDs/DVDs): Plugging an infected USB or external drive can spread malware automatically.
5. Unpatched Software & Exploits: Hackers love outdated software because security holes make it easy to sneak in and attack the system.
6. Malicious Ads (Malvertising): Even legit websites can be compromised by third-party partners that displays ads that can silently install malware without your knowledge.
7. Social Engineering & Human Error: Sometimes, people unknowingly help hackers by sharing sensitive information or following misleading instructions.

Impact of malware include:

- Data Loss: Critical files can be corrupted, encrypted, or deleted.
- Financial Loss: Ransom payments, fraudulent transactions, or repair costs.
- Operational Disruption: Systems and networks can be rendered inoperable.
- Reputational Damage: Loss of customer trust and business credibility.
- Legal Consequences: Failure to protect data can result in regulatory fines and lawsuits.

Some prevention strategies you can employ to protect your computer and data include:

1. Use Antivirus and Anti-malware Software: Regularly update and scan your system with reputable security software.
2. Regular Updates and Patches: Keep your operating system, software, and firmware up to date to close vulnerabilities.
3. Firewalls: Implement hardware and software firewalls to block unauthorized access.
4. Educate Users: Train users to recognize phishing attacks, suspicious links, and unsafe practices.
5. Email Filtering: Use email filters to block malicious emails and attachments.
6. Backup Data: Regularly back up critical data to restore systems in case of an attack.
7. Use Strong, Unique Passwords: Implement complex passwords and change them regularly.
8. Least Privilege Principle: Grant users the minimum level of access necessary to perform their duties.
9. Network Segmentation: Divide your network into segments to limit the spread of malware.
10. Monitor and Audit: Continuously monitor systems for unusual activity and regularly audit security measures.

Computer Ergonomics

First, what is ergonomics?

Ergonomics is the science of designing work environment, tools, and tasks to fit human capabilities and

limitations. It focuses on improving efficiency, comfort, and safety to reduce strain and prevent injuries.

Computer ergonomics is a branch of ergonomics that deals with creating a comfortable and efficient

workspace for computer users. It aims to prevent strain, discomfort, and injuries caused by prolonged

computer use, such as:

- Repetitive strain injuries (RSI) (e.g. carpal tunnel syndrome)
- Eye strain and fatigue
- Neck, back, and shoulder pain
- Poor posture-related issues

Key aspects of computer ergonomics include:

1. Proper Seating & Posture: Use a chair with a good lumbar support to maintain the natural curve

of the spine. The chair height should be adjustable so that the feet are flat on the floor, and knees

are at 90-degree angle. Sit with back straight and shoulders relaxed to avoid hunching.

2. Monitor Placement: The top of the screen should be at eye level to prevent neck strain. Adjust the screen brightness and contrast to reduce eye strain.

3. Keyboard & Mouse Positioning: Keep the keyboard at elbow level to maintain a natural wrist position. Place the mouse close to the keyboard to minimize arm strain.

4. Lighting & Screen Adjustments: Reduce glare by positioning the monitor away from direct light sources. Use blue light filters or night mode to reduce eye strain. Follow the 20-20-20 rule (every 20 minutes, look at something 20 feet away for 20 seconds).

5. Break & Movement: Take frequent breaks to stand, stretch, and move around. Avoid prolonged static postures by adjusting positions regularly.

Computer ergonomics is important, because:

- It prevents long-term health issues
- It improves productivity and focus by reducing discomfort
- Reduces workplace injuries related to prolonged computer use.
- Enhances overall well-being by promoting better posture and movement.

Week 10 & 11: Computer Network

What is a Computer Network?

A computer network is a group of interconnected devices (such as computers, servers, routers, switches,

and other hardware) that can communicate and share resources (such as data, applications, and hardware

components) with each other. These devices are linked using communication channels such as cables,

wireless signals, or fibre optics. These networks can be set in various configurations and sizes, ranging

from small Local Area Networks (LANs) to extensive Wide Area Networks (WANs) that spans large geographical areas.

Importance of Computer Networks

Computer networks play a crucial role in modern computing. Here are some of its importance:

- Resource Sharing: Enables sharing of printers, files, internet connections, and other resources.
- Communication: Enables communication through emails, video conferencing, instant messaging, and VoIP (Voice over Internet Protocol).
- Data Sharing and Accessibility: Allows users to access and share data easily, regardless of location.
- Cost Efficiency: Reduces hardware and software cost by enabling centralized storage and processing.
- Scalability: Networks can be easily scaled by adding new devices without significant changes to the existing infrastructure.
- Improved Security and Backup: Provides better security mechanisms, access control, and

automated backups for data protection.

- Automation and Control: Networks enable automation of tasks and centralized control of connected devices, improving productivity and operational efficiency.

Types of Network Topologies

Network topology refers to the arrangement of devices and how they are connected in a network. The

common types include:

- Bus Topology:** In this topology, all devices are connected to a single central cable (bus). Devices are easy to install and require less cable. The disadvantage of using this topology is, a single cable failure can bring down the entire network.
- Star Topology:** In this topology, all devices are connected to a central hub or switch. Devices are easy to manage, scale, and failure of one device does not affect other devices. The disadvantage of using this topology is that if the central hub fails, the entire network goes down.
- Ring Topology:** In this topology, devices are connected in a circular loop. The data flow in one direction, reducing the risk of data collision. The disadvantage of this topology is that a single point of failure can break the network.
- Mesh Topology:** In this topology, each device is directly connected to every other device. It is highly reliable with multiple paths for data to flow. The disadvantage of this topology is that it is expensive and complex to set up.
- Tree Topology:** This topology forms a structure where devices are connected hierarchically. It resembles a tree with a root node and various branches. The root node is connected to multiple levels of child nodes, forming a hierarchy. The disadvantage of this topology is that if a node fails, the branches will go down too.
- Hybrid Topology:** This topology is a combination of two or more different topologies. It is flexible and scalable based on network requirements. The disadvantage of this topology is that it can be complex and expensive to maintain.

Types of Networks

Networks can be categorized based on their geographical coverage and purpose:

Personal Area Network (PAN)

A Personal Area Network (PAN) is a small network that connects devices within an individual's workspace,

usually within a range of a few meters. It is used for communication between personal devices such as

smartphones, laptops, tablets, and smartwatches.

Characteristics:

- **Short-range Connectivity:** Covers a small area, usually within 10 meters. Primarily connects to an individual's personal devices.
- **Low Power Consumption:** Designed for personal devices with limited power.
- **Wireless or Wired:** Can use Bluetooth, infrared, USB cables, or Wi-Fi
- **Limited Security Risks:** Since it operates in a small area, it has minimal external threats.

Uses:

- Connecting personal devices like smartphones, laptops, and smartwatches.
- File sharing between nearby devices.
- Wireless connection, such as Bluetooth keyboards, mice, and printers.

Local Area Network (LAN)

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area, such as a home, office, or a small group of buildings (e.g., a school or a university campus).

Characteristics:

- Geographic Scope: Limited to a single building or a small group of buildings.
- Speed: Typically, high speed, ranging from 100 Mbps to 10 Gbps.
- Topology: Common topologies include star, ring, and bus.
- Ownership: Usually owned, managed, and maintained by a single organization or individual.
- Media: Often uses Ethernet cables (e.g., Cat 5e, Cat 6) or wireless technologies (e.g., Wi-Fi).
- Cost: Relatively low setup and maintenance costs.

Uses:

- Connecting personal computers, printers, and other devices within an office or home.
- Enabling file sharing, internet access, and peripheral sharing (e.g., printers).
- Supporting internal communications such as email and messaging systems.

Metropolitan Area Network (MAN)

A Metropolitan Area Network (MAN) is a network that covers a larger geographic area than a LAN but smaller than a WAN, typically spanning a city or a large campus.

Characteristics:

- Geographic Scope: Spans a city or a large campus area.
- Speed: High-speed connections, often in the range of 1 Gbps to 100 Gbps.
- Topology: Can include various topologies such as ring, star, or mesh.
- Ownership: May be owned and operated by a single organization or a consortium of users; sometimes leased from service providers.
- Media: Uses fiber optic cables, microwave links, and sometimes wireless technologies.
- Cost: Higher cost compared to LAN due to larger coverage area and infrastructure.

Uses

- Connecting multiple LANs within a city to form a cohesive network.
- Supporting municipal services like emergency response systems, traffic control systems, and public internet access.
- Facilitating communication and data exchange between various branches of an organization within a city.

Wide Area Network

A Wide Area Network (WAN) is a network that spans a large geographic area, often a country or continent, connecting multiple smaller networks, including LANs and MANs.

Characteristics:

- **Geographic Scope:** Can cover large regions, countries, or even continents.
- **Speed:** Variable speeds depending on the technology and infrastructure, ranging from 56 Kbps (dial-up) to several Gbps (fiber optics).
- **Topology:** Often uses mesh, star, or hybrid topologies.
- **Ownership:** Typically owned and operated by multiple organizations or service providers; often involves leasing services from telecom companies.
- **Media:** Utilizes a variety of transmission media, including fiber optic cables, satellite links, and undersea cables.
- **Cost:** High setup and operational costs due to extensive infrastructure and long-distance data transmission.

Uses:

- Connecting geographically dispersed offices and branches of an organization.
- Facilitating global communication and data transfer.
- Supporting the infrastructure of the Internet, which is the largest WAN.
- Enabling remote access to organizational resources and services.

Virtual Private Network (VPN)

A Virtual Private Network (VPN) is a secure network connection that enables users to access the internet

or private networks safely by encrypting data and masking their IP addresses. It is commonly used to

enhance security, privacy, and remote access to corporate networks.

Characteristics:

- **Encryption:** Data is securely encrypted to protect against cyber threats.
- **Remote Access:** Allows users to securely access private networks from anywhere.
- **Anonymity:** Hides the user's real IP address, improving privacy.
- **Bypasses Geo-Restrictions:** Enables access to restricted content in different regions.

Uses:

- Secure remote work by enabling employees to access company networks safely.
- Protecting online privacy by masking IP addresses and encrypting data.
- Accessing restricted content by bypassing regional internet restrictions.
- Preventing cyber threats when using public Wi-Fi networks.

Various LAN Components

1. **Network Interface Cards (NICs):** A network interface card (NIC) is a hardware component that is used to connect computers and devices to a network and allow them to communicate. It can be built into the motherboard or installed as an expansion card. NICs can be wired (Ethernet) or Wireless (Wi-Fi).

2. Switch: A switch is a device that receives, processes, and forwards data to the destination device on the same network. It operates at the data link layer (Layer 2) of the OSI model. It uses MAC (Media Access Control) addresses to forward data efficiently.
3. Hub (Legacy Component): A hub is an outdated networking device that broadcast data to all connected devices. Unlike switches, hubs do not filter traffic, leading to network congestion.
4. Router: A router is a device that directs data packets between networks, such as between a LAN and the Internet. It operates at the network layer (Layer 3) of the OSI model. Routers can be Wired routers and wireless routers, with capabilities like Firewall, Network Address Translation (NAT), and Dynamic Host Configuration Protocol (DHCP).
5. Access Point (AP): An access point is a device that allows wireless devices to connect to a wired network using Wi-Fi. It connects to a switch or router to extend the network wirelessly. It acts as a bridge between the wired and wireless segments of the network.
6. Cables and Connectors: Various types of cables are used in LANs to connect devices, such as Ethernet cables for wired connections and fiber optic cables for high-speed connections. RJ-45 connectors are used to terminate Ethernet cables.
7. Servers: A server is a device that provides network services to other computers (clients) on the network, such as file storage, printing, or authentication. Common types of servers include file servers, print servers, database servers.
8. Clients: Clients are the computers and devices that users interact with. They connect to the network to use shared resources and services. Examples are Desktop computers, laptops, smartphones, and tablets.
9. Network Operating System (NOS): A network operating system is specialized software that manages network resources, users, and devices. It provides services like file sharing, printer access, user management, and security. Examples include Microsoft Windows Server, Linux-based systems, and Novell NetWare.
10. Firewall: A firewall is a security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules. It protects the LAN by filtering incoming and outgoing traffic. It can be hardware-based (part of a router) or software-based.
11. Modem: A modem is used to connect a LAN to an Internet Service Provider (ISP) like MTN, Glo, for internet access.

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Level ND1 • First Semester • Handout

Week 1 & 2: History, Classification & Impact of Computers

What is a Computer?

A computer is an electronic device that processes data and perform tasks according to a set of instructions

called a program to produce meaningful information. It can store, retrieve, and manipulate data to produce useful information.

Computers can perform a variety of tasks, including calculations, word processing, communication, and

data storage. Computers are used in almost every field today, from education and healthcare to entertainment and business.

Basic Elements of a Computer System

A computer system consists of four main elements:

1. Hardware: The physical parts of the computer. The hardware is divided into:

- Input Devices: Devices used to enter data into the computer, e.g. keyboard, mouse, microphone.
- Output Devices: Devices that display or present information from the computer, e.g. monitor, printer, speakers.
- Processing Unit: The Central Processing Unit (CPU), which acts as the brain of the computer, executing instructions.
- Memory (RAM and Storage): Temporary memory (RAM) stores data while the computer is on, while permanent storage (e.g. hard drive, SSD) saves data long-term.

2. Software: The programs and applications that run on the hardware, such as operating systems (Windows, macOS) and apps (Microsoft Word, Call of Duty, Google Chrome). The computer software is divided into two:

- System Software: These are software that are responsible for managing hardware and creating an environment where application software can run. It works at a fundamental level and ensures that the computer functions properly. Examples of system software are: operating system, device drivers, utility software, firmware.
- Application Software: These are software designed for end-users to perform specific tasks, such as word processing, web browsing, entertainment, etc. Examples of application software are: productivity software (Microsoft Word, Excel, PowerPoint), web browser (Google Chrome, Mozilla Firefox, Microsoft Edge), multimedia software (Adobe Photoshop, VLC Media Player), communication software (WhatsApp, Zoom, Twitter), entertainment software (Spotify, Call of Duty, Netflix).

3. Data: The raw facts and figures that the computer processes to produce meaningful information.

Data can be:

- Text Data: Letters, numbers, and characters typed into a document.
- Numerical Data: Numbers used in calculations, such as financial records.
- Image Data: Pictures, graphics, and scanned documents.
- Audio Data: Sounds and voice recordings.
- Video Data: Moving pictures and animations.

4. Users: The people who interact with the computer to perform tasks.

History of Computers

Primitive humans used the first counting device. They counted with sticks, stones, and bones. As technology and human intellect advanced over time, more computing devices were created. Some of the

popular computing devices starting with the first to recent ones are described below;

Abacus

The abacus, which is thought to have been the earliest computer, is credited with starting the history of

computers. Abacus is thought to have been created by the Chinese some 4,000 years ago.

The wooden rack has metal rods with beads attached to them. In order to do mathematical computations,

the abacus operator moved the beads in accordance with certain principles. Several nations, including

China, Russia, and Japan, continue to use abacuses.

Napier's Bones

This manually driven calculator was created by Merchiston resident John Napier (1550-1617). He multiplied and divided using nine distinct ivory strips or bones that were numbered in this calculator. Thus,

the instrument was dubbed "Napier's Bones." Additionally, it was the first device to work with the decimal point.

Pascaline

Pascaline is sometimes referred to as an adding machine or an arithmetic machine. It was created by the

French mathematician and philosopher Blaise Pascal between 1642 and 1644. It is thought to have been

the first automatic and mechanical calculator.

In order to assist his father, a tax accountant, Pascal created this device. It was limited to adding and

subtracting. It was a wooden box with wheels and a number of gears. A wheel turns the wheel next to it

when it completes one rotation. To read the totals, a number of windows are provided on top of the wheels.

Leibnitz Wheel

It was created in 1673 by Gottfried Wilhelm Leibnitz, a German mathematician and philosopher. To create

this machine, he modified Pascal's idea. Because it used stepped drums in place of gears, this digital

mechanical calculator was known as the "stepped reckoner."

Difference Engine

In the early 1820s, Charles Babbage—dubbed the "Father of Modern Computer"—designed it. It was a

mechanical computer with basic computation capabilities. It was a steam-powered calculator made to

solve numerical tables, such as logarithm tables.

Analytical Engine

Additionally, Charles Babbage created this calculator around 1830. Punch cards were utilized as input for

this mechanical computer. It has the ability to store data in a permanent memory and solve any mathematical problem.

Tabulating Machine

It was created in 1890 by American statistician Herman Hollerith. Punch cards served as the basis for this

mechanical tabulator. In addition to recording or sorting data or information, it could tabulate statistics.

The U.S. Census of 1890 made use of this device. The Hollerith's Tabulating Machine Company, which

Hollerith founded, changed its name to International Business Machine (IBM) in 1924.

Differential Analyzer

In 1930, the United States saw the introduction of the first electronic computer. Vannevar Bush created

the analogue gadget. This device uses vacuum tubes to switch electrical signals so that computations can

be done. It took only a few minutes to complete 25 calculations.

Mark I

The next significant developments in computer history started in 1937 when Howard Aiken aimed to

create a device capable of carrying out complex mathematical computations. IBM and Harvard collaborated to build the Mark I computer in 1944. It was the first digital computer that could be programmed.

Generations of Computers

The history of computer development is often referred to as the different generations of computing

devices. Each generation of computers is characterized by a major technological development that fundamentally changes the way computer operates, resulting in an increasingly smaller, cheaper, more

efficient and reliable devices. It replaced the gears and other mechanical parts used for counting in previous computing machines. The different generations of computing devices are:

1. First Generation Computers (1940 – 1956): The first computers used vacuum tubes for circuitry and magnetic drums for memory. They were often too huge, occupying entire rooms, very expensive to operate, used a great deal of electricity, and generated a lot of heat which was often the cause of malfunctions. First generation computers relied on machine language to perform operations, and they could only solve one problem at a time. Input was based on punch cards and paper tapes, and output was displayed on printouts. Example of first generation computers are UNIVAC (Universal Automatic Computer) and ENIAC (Electronic Numerical Integrator and Computer).

2. Second Generation Computers (1956 – 1963): Transistors replaced vacuum tubes and ushered in the second generation of computers. The transistor was invented in 1947, but did not see widespread computer use until the late 50s. The transistor was far superior to the vacuum tubes, allowing computers to become smaller, faster, cheaper, more energy efficient, and more reliable than their first generation predecessors. Though the transistors still generated a great deal of heat that subjected the computer to damage, it was a vast improvement over the vacuum tubes. Second generation computers still relied on punched cards for input and printouts for output. Second generation computers moved from cryptic binary machine language to symbolic or assembly languages, which allowed programmers to specify instructions in words. High level programming languages were also being developed at this time such as early versions of COBOL and FORTRAN. These were also the first computers that stored their instructions in their memory which moved from a magnetic drum to a magnetic core technology. The first computers of this generation were developed for the atomic energy industry.

3. Third Generation Computers (1964 – 1971): The development of the integrated circuit was the hallmark of the third generation of computers. Transistors were miniaturized and placed on silicon chips called semi-conductors which drastically increased the speed and efficiency of computers. Instead of punched cards and printouts, users interacted with third generation computers through keyboards and monitors, and interfaced with an operating system which allowed the device to run many different applications at one time with a central program that monitored the memory. Computers for the first time became accessible to the public because they were smaller and cheaper than their predecessors.

4. Fourth Generation computers (1971): The microprocessor brought the fourth generation of computers as thousands of integrated circuits were built into a single silicon chip. What in the first generation filled an entire room could now fit in the palm. The Intel 404 chip developed in the 1971 located all the components of the computer from the central processing unit and memory to input-output controls on a single chip. In 1981, IBM introduced its first computer for the home user and in 1984 Apple introduced the Macintosh. Microprocessors also moved into many areas of life as more and more everyday products began to use microprocessors, as these small computers became more powerful, they could be linked together to form networks which eventually led to the development of the internet. Fourth generation computers also saw the development of GUIs (Graphical User Interfaces), the mouse, and handheld devices.

5. Fifth Generation Computers (present): Fifth generation computing devices, based on artificial

intelligence are still in development, though there are some applications such as voice recognition that are being used. The use of parallel processing and superconductors is helping to make artificial intelligence a reality. The goal of this generation computing is to develop devices that respond to natural language input and are capable of learning and self-organization.

Classification of Computers

Computers can be classified based on their size, mechanism, and purpose/use.

Classification of computer

Based on Size Based on Mechanism Based on Purpose/Use

- ✓ Super computer ✓ Analog computer ✓ General purpose
- ✓ Mainframe computer ✓ Digital computer ✓ Special purpose
- ✓ Mini computer ✓ Hybrid computer
- ✓ Micro computer
- Desktop
- Laptop
- Palmtop

Super Computer

- They are very fast and powerful machines.
- Run at the speed of 10 MIPS (million instructions per second).
- Very expensive.
- Not used for general applications.
- Examples: Cray and CDC Cyber.

Mainframe Computer

- They are built for general computing (i.e. can be used for business or engineering).
- Very fast and will process information at about 10 MIPS. They have large storage capacity.
- Very expensive.
- Not easily found in general design offices.
- They can handle 100s – 1000s of users at the same time.

Mini Computer

- Developed in the 1960's resulting from advances in microchip technology.
- Smaller and less expensive than mainframe computers.
- Run at several MIPS and can support 5-20 users.
- Low cost and high performance.
- Used in small organizations.

Micro Computer

- Invented in the 1970's.
- Generally used for home and personal computing.
- Dedicated data processing workstations.
- Handle one user at a time.
- They are cheap.

Desktop computer is a personal computer designed to fit on a small office desk. It includes: monitor, keyboard, mouse, system unit, etc.

Laptop is a personal computer designed for mobility. It is small and light to place on the lap of the user. It integrates most of the components of the desktop.

Palmtop is a small computer that literally fits on your palm. Due to their small size, they are limited in capabilities.

Analog Computer

- The word <analog= means continuously changeable in quantity.
- The analog computers accept input data in continuous form and output is obtained in the form of graphs. Example: voltage, current, sound, speed, temperature, pressure, etc.
- The analog computers are used to measure the continuous values. Example: thermometer, speedometer, voltmeter, clock, etc.

Digital Computer

- The word <digital= means separate. It refers to binary system, which consists of only two discrete digits, i.e. 0 and 1.
- Digital data consists of binary data represented by OFF (low) and ON (high) electrical pulse.
- In digital computers, quantities are counted rather than measured.
- A digital computer operates by counting numbers or digits and give output in digital form. Examples: calculators, personal computers, digital watches, digital thermometer.
- They are used in offices, education, hotels, hospitals, data processing, finance, etc.

Hybrid Computer

- They combine the feature of both analog and digital computers.
- In hybrid computers, the user can process both the continuous (analog) and discrete (digital) data by converting from one form to another using analog-to-digital converter (ADC) and digital-to-analog converter (DAC).
- These are special purpose computers, such as a machine that measures patients heart beat and convert the analog data into digital form to display on the screen, or dispenser machines at petrol pumps that converts the measurements of fuel flow into quantity and price.
- They are very fast and accurate.
- These are used in scientific fields such as weather forecasting, airline systems, etc.

Special Purpose Computer

- These are designed to perform a specific task. Such computers lack flexibility.
- They perform the task for which they are designed very efficiently.
- Examples: air craft control system, missile guidance system, heart beat monitor, etc.

General Purpose Computer

- A general purpose computer is one that can be used for a variety of applications.
- Its creativity enables execution of programs of almost any type. These are used in business

applications such as word processing, spreadsheet, database, games, communications, etc.

Benefits of Computers to the Society

Computers have become an integral part of modern society, transforming many aspects of our lives with

their versatility and efficiency.

The benefits of computers span many different sectors, including education, business, healthcare, and

personal development. Understanding these benefits highlights the integral role computers play in shaping an advanced and connected world.

1. Education

- E-learning platforms enable online learning.
- Digital libraries provide access to research materials.
- Examples: Coursera, Udemy, YouTube, Khan Academy

2. Communication

- Emails, instant messaging, and video calls connect people worldwide.
- Examples: WhatsApp, Zoom, Gmail.

3. Healthcare

- Computers help in diagnosing diseases, managing patient records, and conducting surgeries.
- Examples: MRI Scanners, Electronic Medical Records (EMR).

4. Business and Finance

- Automate tasks such as payroll, accounting, and stock trading.
- Examples: Banking software, E-commerce platforms like Amazon, Jumia.

5. Transportation

- GPS systems help navigation.
- Airline and railway booking systems use computers for scheduling.
- Examples: Google Maps, Air Traffic Control Systems.

6. Entertainment

- Computers are used in gaming, movie production, and music streaming.
- Examples: Netflix, Spotify, PlayStation.

Social Implications of Computers on Society

While computers bring many benefits, they also have significant social implications, particularly in privacy

and quality of life.

Privacy Concerns

- Personal data is collected by various websites, social media, and apps, raising concerns about data security.
- Cybercrime (hacking, phishing, identity theft) has increased.
- Governments and companies monitor online activities (e.g. surveillance systems).

Impact on Quality of Life

- Social isolation due to excessive screen time.
- Online communication have changed how people interact, with virtual relationships replacing real-life interactions, leading to social isolation.
- Health issues like eye strain, obesity, and poor posture due to prolonged computer use.
- Job losses due to automation replacing human labor.

Digital Divide

- Not everyone has equal access to computers and the internet, leading to disparities in education, employment, and communication.
- Wealthier individuals and developed countries benefit more from technology than poorer regions.

Ethical Issues

- The rise of AI and automation raises ethical concerns about surveillance, deepfakes, and bias in decision-making systems.
- Misuse of technology for illegal activities (e.g. cyberbullying, spreading fake news).

Environmental Impact

- The production and disposal of computers contribute to electronic waste (e-waste).
- Data centers consume large amounts of energy, increasing carbon footprints.

Week 3: Computer Hardware

Introduction to Computer Hardware

Computer hardware refers to the physical components of a computer system that you can see, touch, and

interact with. These components work together to perform tasks, process data, and display output.

Without hardware, a computer cannot function.

Types of Computer Hardware and their Functions

Hardware components can be classified into different types based on their roles in the computer system:

Input Devices

Input devices allow users to enter data and instructions into the computer. They serve as a bridge between

humans and computers.

- Keyboard: Used for typing text, numbers, and special characters.
- Mouse: A pointing device used to navigate and interact with objects on the screen.
- Scanner: Converts physical documents or images into digital form.
- Microphone: Captures sound and converts it into digital data.
- Webcam: Captures images and videos for video conferencing or recording.
- Joystick/Game Controller: Used for gaming and controlling movement in virtual environment.

Output Devices

Output devices allow the computer to display, print, or produce information to the user in the desired

format.

- **Monitor (Screen/Display):** Displays visual information from the computer.
- **Printer:** Produces a hard copy of digital documents.
- **Speaker:** Converts digital audio signals into sound.
- **Headphones:** Output sound privately to a single user.
- **Projector:** Displays images or videos on a large screen.

Processing Unit (Central Processing Unit – CPU)

The CPU (Central Processing Unit) is the <brain> of the computer. It processes all instructions and performs

calculations necessary for running applications. The CPU consists of three main components:

- **Control Unit (CU):** Directs the flow of data and instructions within the computer.
- **Arithmetic and Logic Unit (ALU):** Performs mathematical calculations and logical operations.
- **Registers:** Temporary storage areas within the CPU that hold data being processed.

The speed of the CPU is measured in gigahertz (GHz), and a higher GHz value means faster processing power.

Storage Devices

Storage devices are used to store data, programs, and files either temporarily or permanently. They are

categorized into four:

- **Random Access Memory (RAM):** It is the primary memory of the computer that stores data temporarily. It stores data that is currently in use. It is volatile (data is lost when power is off).

Example of such memory are DDR 3, DDR 4, DDR 5 RAM.

- **Read-Only Memory (ROM):** It is a permanent memory that stores essential system instructions. It stores firmware and boot-up instructions (BIOS). It is non-volatile (data remains even when power is off). It is fast but slower than RAM. Examples of such memory are BIOS, firmware chips. You cannot modify the data in ROM, but can only read from it.

- **Secondary Storage:** It is a permanent storage for user data, files, and software. It stores data long-

term, even when the computer is off. It is non-volatile, and is slower than RAM and ROM. Example of such storage are HDD, SSD, USB flash drives.

- **Auxiliary Memory:** Another term for secondary storage; sometimes includes external storage. Provides extra storage for data, backups, and large files. It is non-volatile, and is slower than RAM and ROM. Example of such storage are HDD, SSD, USB flash drives.

Other important hardware components include:

- **Motherboard:** The motherboard is the main circuit board that connects all computer components, including the CPU, RAM, storage, and input/output devices.
- **Power Supply Unit (PSU):** The power supply unit converts electricity from an external source (e.g. NEPA, BEDC, EEDC) into a form that the computer can use.
- **Cooling Systems (Fans & Heat Sink):** Computers generate heat while operating. Cooling Systems, such as fans and heat sinks, prevent overheating by dissipating heat.

Week 4: Central Processing Unit

What is a Central Processing Unit (CPU)?

The Central Processing Unit (CPU), often referred to as a computer processor, microprocessor or simply

processor, is the brain of the computer. It provides processing power to the computer system. It carries

out instructions from software programs by performing basic arithmetic, logic, control, and input/output

operations. Without the CPU, a computer cannot function.

It performs most of the processing operations inside a computer, executing instructions and managing

data manipulation. The CPU fetches and execute instructions from the primary memory (RAM), coordinating the activities of all the hardware components to perform various tasks.

There are two main computer processor manufacturers: Intel and AMD (Advanced Micro Devices). These

two companies produce most of the processors used in desktop, laptops, and notebooks.

A CPU is placed on the mother motherboard in the CPU socket with a locking lever mechanism in order to

fix the CPU properly into the CPU socket. A heat-sink is mounted on top of the CPU chip which protects

the CPU from the excessive heat generated by the CPU. The heat-sink absorbs the heat, and is then cooled

by a fan placed on top of the heat-sink.

Main Components of the CPU

The performance of a CPU is influenced by many factors like architecture, clock speed, cache size, and the

number of cores. The arithmetic logic unit, control unit, and registers are the core parts of the CPU.

Others

are referred to as auxiliary units as they enhance the CPU efficiency by managing memory, speeding up

processing, and handling complex operations.

1. Control Unit: The control unit acts as the manager of the CPU. It directs the flow of data between the CPU and other parts of the computer, telling the ALU and memory how to respond to the instructions it receives. It fetches, decodes, and execute instructions from memory.

2. Arithmetic Logic Unit (ALU): The ALU performs all the mathematical calculations and logical operations. Arithmetic operations include basic calculations like addition, subtraction, multiplication, and division. Logical operations involve comparing values (like determining if one number is greater than or equal to another).

3. Registers: Registers are small, fast storage locations within the CPU that hold data temporarily while the CPU processes it. There are different types of registers, such as:

a. Accumulator (AC): Stores intermediate results during computations.

b. Program Counter (PC): Keeps track of the next instruction to be executed.

- c. Instruction Register (IR): Holds the current instruction being executed.
- d. Memory Address Register (MAR): Holds the address in memory where data is to be read from or written to.
- 4. Cache: The cache is a small amount of high-speed memory that stores frequently accessed data and instructions to speed up processing. It is closer to the CPU than regular RAM, so it allows for quicker access to critical data. There are three levels of cache:
 - a. L1 Cache: The smallest and fastest, directly integrated into the CPU core.
 - b. L2 Cache: Larger but slower than L1, still within the CPU chip.
 - c. L3 Cache: Larger and shared among all CPU cores, slower but larger than L2.
- 5. Clock: The clock in the CPU generates the electrical pulses that synchronize the operations of the CPU. The clock speed is measured in GHz (gigahertz), and it determines how many instructions the CPU can process per second. A faster clock speed generally means a faster CPU.
- 6. Bus Interface: This is responsible for managing the flow of data between the CPU and other components, such as memory or input/output devices. There are different types of buses:
 - a. Data bus: Transfers data between the CPU and other components.
 - b. Address bus: Carries the addresses of where data is to be fetched or sent.
 - c. Control bus: Carries control signals from the control unit to other parts of the computer.
- 7. Instruction Decoder: The instruction decoder interprets the instructions fetched from the memory and converts them into signals that the ALU and other components can understand and act upon.
- 8. Floating Point Unit (FPU): The FPU handles complex arithmetic involving floating-point numbers (decimals). It's separate from the ALU, which primarily handles integer operations.
- 9. Cores: Modern CPUs have multiple cores, which are individual processing units within the CPU. A multi-core CPU can handle multiple tasks simultaneously (parallel processing), significantly improving performance for multitasking or heavy workloads.

How the CPU Works

The CPU operates through a cycle called the fetch-decode-execute cycle.

- Fetch: Retrieves an instruction from the memory.
- Decode: Interprets what the instruction means.
- Execute: Carries out the instruction and sends results to the appropriate part of the computer.

Functions of the CPU

- To perform arithmetic operations.
- To perform logical operations.
- To control functions of other hardware components.
- To fetch the data and program instructions from main memory RAM.
- To decode the program instructions.
- To operate on the data as per program instructions.
- To store the processed data into the RAM.
- To continuously execute instructions.

Types of Computer Memory

- Random Access Memory (RAM): It is the primary memory of the computer that stores data

temporarily. It stores data that is currently in use. It is volatile (data is lost when power is off).

Example of such memory are DDR 3, DDR 4, DDR 5 RAM.

- Read-Only Memory (ROM): It is a permanent memory that stores essential system instructions.

It stores firmware and boot-up instructions (BIOS). It is non-volatile (data remains even when

power is off). It is fast but slower than RAM. Examples of such memory are BIOS, firmware chips.

You cannot modify the data in ROM, but can only read from it.

- Secondary Storage: It is a permanent storage for user data, files, and software. It stores data long-

term, even when the computer is off. It is non-volatile, and is slower than RAM and ROM. Example

of such storage are HDD, SSD, USB flash drives.

- Auxiliary Memory: Another term for secondary storage; sometimes includes external storage.

Provides extra storage for data, backups, and large files. It is non-volatile, and is slower than RAM

and ROM. Example of such storage are HDD, SSD, USB flash drives.

Functions of Auxiliary Memory

Auxiliary memory (also called secondary storage) refers to devices that store data permanently, even

when the computer is turned off. Its functions include:

- Long-term data storage: Saves files, applications, and operating systems.
- Backup and recovery: Protects data from loss due to system failure.
- Data transfer: Moves files between devices using external storage (e.g. USB drives).
- Virtual memory support: Acts as extra storage when RAM is full, helping the system run smoothly.

Measurements of Storage

Computer storage is measured in bytes, with larger units increasing in powers of 1,024.

1. Bit (b): The smallest unit of data representing either 0 or 1.
2. Byte (B): 8 bits make up one byte, enough to store one character (e.g. <A=, <1=, <@=).
3. Kilobyte (KB): 1,024 bytes (approximately 1,000 bytes).
4. Megabyte (MB): 1,024 KB (approximately 1 million bytes).
5. Gigabyte (GB): 1,024 MB (approximately 1 billion bytes).
6. Terabyte (TB): 1,024 GB (approximately 1 trillion bytes).
7. Petabyte (PB): 1,024 TB (approximately 1 quadrillion bytes).
8. Exabyte (EB), Zettabyte (ZB), Yottabyte (YB): Extremely large storage capacities used in data centers and cloud storage.

Week 5: Concept of Computer Software

What is Computer Software?

Computer software is basically a program that tells the computer's hardware what to do.

First let's understand that hardware devices need user instruction to function. A set of instructions that

achieve a single outcome are called a program or procedure.

So, a computer software can be defined as a collection of programs, data, and instructions that tell a

computer how to perform specific tasks. Unlike hardware, which refers to the physical components of a

computer, software consists of code and digital instructions that enable hardware to function.

Types of Software

Computer software is broadly categorized into two main types: system software and application software.

User

Application Software

System Software

Computer Hardware

System Software

System software is a software that controls a computer's internal functioning, through an operating

system, and also controls such peripherals such as monitors, printers, and storage devices. It is also responsible for creating an environment where application software can run. It works at a fundamental

level and ensures that the computer functions properly.

Characteristics of system software include:

- Runs in the background and starts when the computer is powered on.
- Directly interacts with the hardware components (CPU, memory, storage, etc).
- Provides a platform for users to install and run application software.
- Typically developed using low-level programming languages like C and Assembly.

Types of System Software

1. Operating System (OS): This is the most important system software that manages all computer resources. It is usually the first software to be loaded into the computer memory. It acts as an intermediary between the user of a computer, application software and the computer hardware. It manages the computer's basic functions like storing data in memory, retrieving files from storage devices, scheduling tasks based on priority, etc. Examples: Windows, MacOS, Linux, Android & iOS (mobile).

2. Firmware: This is a low-level software that's embedded in hardware like ROM chips to control basic functions. Examples are:

- a. BIOS (Basic Input/Output System): This manages the startup settings before the OS loads.
- b. UEFI (Unified Extensible Firmware Interface): The modern replacement for BIOS. Provides better encryption and GUI to navigate.

3. Device Driver: This is a special software that enables the operating system to communicate with hardware devices. It controls and monitors functioning of a specific device on the computer. Each device attached to the computer has a specific driver associated with it. When you attach a new device, you need to install its driver so that the OS knows how to manage the device. Examples: printer driver, graphics driver, sound driver.

4. Utility Software: Helps maintain the system and improve performance. Examples: antivirus software, disk management tools, file management tools, compression tools, backup tools.

Application Software

Application software is a software designed for end-users to perform specific tasks, such as word processing, web browsing, entertainment.

Characteristics of application software include:

- Runs on top of system software.
- Designed for a specific function like editing documents, playing games, browsing the internet.
- Can be installed, updated, and removed as needed.
- Developed using high-level programming languages like Python, Java, C++, etc.

Types of Application Software:

1. **Productivity Software:** Used for office and business tasks. Examples: Microsoft Suite (Word, Excel, PowerPoint, Access), Google Workspace (Gmail, Docs, Sheets, Slides).
2. **Web Browsers:** Allows the user to access and navigate the internet. Examples: Google Chrome, Mozilla Firefox, Microsoft Edge.
3. **Multimedia Software:** Used for creating, editing, and playing media. Examples: Adobe Photoshop, VLC media Player.
4. **Communication Software:** Enables users to communicate via text, voice, video. Examples: WhatsApp, Zoom, Skype.
5. **Entertainment and Gaming Software:** Used for leisure and gaming purposes. Examples: Spotify, Netflix, Call of Duty.
6. **Educational Software:** Helps users learn new skills or subjects. Examples: Duolingo, Khan Academy, MATLAB.

Key differences between system software and application software.

Feature	System Software	Application Software
Purpose	Manages the hardware and provides a platform for applications	Helps users perform specific tasks
User Interaction	Runs in the background; users rarely interact with it directly	Users interact with it directly
Dependency	Needed for the computer to function	Runs on system software
Examples	Operating Systems, Drivers, Utility	Microsoft Word, Chrome, Photoshop
Execution	Starts when the computer boots and runs continuously	Starts when opened and closes when exited

Computer Languages

Computer language is the language used to communicate with the computer.

Computer language is commonly referred to as programming language. A programming language is a set

of rules, symbols, and syntax used to write instructions that a computer can understand and execute.

People who write in this language are called programmers. This allows programmers to create software,

applications and systems by providing the commands the computer should follow.

Computer language is divided into two: low-level language and high-level language.

Low-Level Languages

These languages interact directly with the hardware and are faster but harder to understand. There are

two types of low-level programming languages:

- **Machine Language:** Computers can understand only digital signals, which are in binary digits (0 and 1). So, the instruction given to the computer can be only in binary codes. This is the only language the computer directly understands. Writing a program in machine level language is a difficult task because it is not easy for computer to write instructions in binary codes. There is no need for any translator, since the computer is already communicating in the machine level language. Example: 01000001 01000010.

- **Assembly Language:** The difficulties faced in machine level language were reduced to some extent

by using a modified form of machine level language called assembly language. In assembly language, instructions are given in English like words (mnemonics) such as MOV, ADD, SUBB, etc instead of raw binary. Assembly language is hardware-dependent, meaning it is not easily portable across different types of computers. Assembly language is translated into machine level language using a translator known as <assembler>. Example: MOV A, B (moves data from register B to A).

Assembly Machine

Assembler

Language Language

High-Level Language

High-level language is a programming language that is written in human readable language. It is easier

to read, write, and maintain. It is portable, and not designed to run on just one type of machine. A high-

level language is translated into a machine language using a compiler or an interpreter. Example of high-

level language are Python, Java, C++, JavaScript.

High-Level Machine

Compiler/Interpreter

Language Language

Source and Object Programs

In programming, when we write code, it needs to be converted into a form that the computer can understand. This process involves two key terms: source program and object program.

Source Program

A source program is the original code written by a programmer in a high-level language or assembly

language. This code is written in a human-readable format and must be converted into machine language

before execution. This program can be modified and debugged easily.

Object Program

An object program is the translated version of a source program, written in machine code (binary), which

the computer can execute. It is generated by an assembler, compiler, or interpreter from the source code.

The object program is not human readable, but is understandable by the computer, hence it can be executed directly by the computer. It can be linked with other object files to form an executable file.

So, we can say the source program is what the programmer writes. And the object program is the machine-code version of the source program, which the computer can execute.

Language Translators

A translator is a program or processor that converts code written in one programming language into another language, typically from a high-level or low-level language into machine code that the computer

can execute.

There are three main types of translators:

- Assembler
- Interpreter
- Compiler

Each type of translator works differently and is used for specific types of programming languages.

Assembler

Assembler is a translator program that is used to convert an assembly language into a machine language.

The assembler converts the assembly language mnemonics into machine code that the CPU can execute.

They are fast in translation since it works with low-level instructions. The output is hardware-specific,

meaning different processors require different assembly languages.

Interpreter

Interpreter is a translator program that is used to convert high-level language program into a machine

language. It translates the program line by line rather than all the code at once. If there is an error in any

line, the execution stops at the line. It has slower execution time because translation happens during

runtime. Also, it produces no separate executable file; the source code must be interpreted every time it

runs.

Compiler

Compiler is a translator program that is used to convert high-level language program into a machine

language. It translates the whole program at once and generates the object code or file with a list of errors

if present. The object file is then linked with libraries to produce an executable file (.exe). The computer

can run the executable file without needing the original source code.

Computer Packages and user Application Software

Both computer packages and user application software refer to programs designed to perform specific

tasks on a computer. However, they serve different purpose and are used in different contexts.

Computer Packages

A computer package is a set of pre-designed software programs that work together to perform specific

functions. These packages are usually bundled and provide a complete solution for users in various industries. Examples:

Category Examples Purpose

Office Suite Microsoft Office, Google workspace Word processing, spreadsheet, presentations

Graphics and Design Adobe creative suite, CorelDraw Photo editing, graphic design, video editing

Database Microsoft access, Oracle, MySQL Storing and managing large data

Management

Statistical and SPSS, MATLAB, R Data analysis, research, and

Scientific engineering simulations.

User Application Software

User application software refers to individual programs designed to help users perform specific tasks on

a computer. These can be standalone or part of a larger software package.

Category Examples Purpose

Word processing Microsoft Word, Google docs Creating and editing documents

Spreadsheet Microsoft Excel, Google Sheets Data analysis and financial calculation

Web browser Google Chrome, Mozilla Firefox Accessing the internet

Week 6: Data Processing

What is Data Processing?

Data in its raw form is not useful to any organization. Data processing is the method of collecting raw data

and translating it into usable information. It involves collecting, organizing, analyzing, storing, and presenting data in a way that is useful for decision-making.

In simpler terms, it is the method by which raw facts (data) are converted into meaningful outputs

(information) through computational or manual operations.

Data Processing Cycle

The data processing cycle consists of a series of steps where raw data (input) is fed into a system to produce actionable insights (output). Each step is taken in a specific order, but the entire process is repeated in a cyclic manner. The first data processing cycle's output can be stored and fed as the input for the next cycle, as the illustration below shows us.

Fig: Data processing cycle

Generally, there are six main steps in the data processing cycle:

Step 1: Collection

The collection of raw data is the first step of the data processing cycle. The type of raw data collected has a huge impact on the output produced. Hence, raw data should be gathered from defined and accurate sources so that the subsequent findings are valid and usable. Raw data can include monetary figures, website cookies, profit/loss statements of a company, user behavior, etc.

Step 2: Preparation

Data preparation or data cleaning is the process of sorting and filtering the raw data to remove unnecessary and inaccurate data. Raw data is checked for errors, duplication, miscalculations or missing data, and transformed into a suitable form for further analysis and processing. This is done to ensure that only the highest quality data is fed into the processing unit.

The purpose of this step is to remove bad data (redundant, incomplete, or incorrect data) so as to begin assembling high-quality information so that it can be used in the best possible way for business intelligence.

Step 3: Input

In this step, the raw data is converted into machine readable form and fed into the processing unit. This can be in the form of data entry through a keyboard, scanner or any other input source.

Step 4: Data Processing

In this step, the raw data is subjected to various data processing methods using machine learning and artificial intelligence algorithms to generate a desirable output. This step may vary slightly from process to process depending on the source of data being processed (data lakes, online databases, connected devices, etc.) and the intended use of the output.

Step 5: Output

The data is typically transmitted and displayed to the user in a readable form like graphs, tables, vector files, audio, video, documents, etc. This output can be stored and further processed in the next data processing cycle.

Step 6: Storage

The last step of the data processing cycle is storage, where data and metadata are stored for further use.

This allows for quick access and retrieval of information whenever needed, and also allows it to be used as input in the next data processing cycle directly.

Types of Data Processing

Data processing techniques refer to the different ways data can be handled or processed based on the

mode of operation. They define how data flows and is transformed. It focuses on when and how data is

structured and scheduled for processing – whether all at once (batch), instantly (real-time), across multiple

computers (distributed), or shared among users (time-sharing).

Batch Processing

In batch processing, data is collected over a period and processed at a later time as a single unit. It is used

for tasks that do not require immediate updates.

- It is cost effective for large data sets
- Can be scheduled (e.g. nightly, weekly processing)
- Suitable for repetitive tasks

Examples:

- Payroll Processing: Employee salaries are calculated and paid at the end of the month.
- Bank Transaction Processing: Banks process multiple transactions in batches at night.
- Sales Report Processing: A supermarket collects sales data throughout the day and processes all transactions at night.

Real-Time Processing

In real-time processing, data is processed immediately as it is received, without delay. It is commonly used

in applications where quick decision-making is required.

- It is extremely fast
- Processes data as events occur
- Requires high computing power

Examples:

- ATMs: Updates account balances immediately after a withdrawal.

- Stock Market Trading: Processes transactions in milliseconds.
- Online Gaming: Real-time game data updates for players worldwide.
- Inventory Management System: The cash register updates inventory immediately customer make purchases.

Online Processing

In online processing, data is stored in a database and is processed over a network, often in real time or later in batches.

- High speed and accuracy
- Can handle large datasets
- Uses modern computing technologies like databases, cloud computing, and artificial intelligence

Examples:

- Online Banking: Customers account balances are updated in real time as you transfer money.
- Online Stores: An online grocery store updates stock levels as customers place orders.

Time-Sharing Processing

In time-sharing processing, multiple users or processes share computing resources simultaneously. The

system allocates small time slots (CPU time slices) to each task in a rotating manner. This method allows

multiple users to interact with a computer system simultaneously without significant delays.

- Many users can access and use the system at the same time
- The processor switches between tasks rapidly, ensuring fair usage
- The system responds almost instantly to user inputs
- Maximizes the use of CPU and memory

Examples:

- Online Banking Systems: Multiple users log into the bank's website at the same time. The server assigns processing time to each user request, ensuring smooth transactions.
- Airline Reservation Systems: Hundreds of users book flights online. The system processes each request in time slices, preventing long wait times.
- Cloud Computing Services: Many users access and edit documents stored in the cloud at the same time. The server distributes processing power to handle multiple requests efficiently.

Distributed Processing

In distributed processing, multiple computers work together to process data across different locations. It

is commonly used in cloud computing and large organizations.

- Increases efficiency by sharing workloads
- Reduces processing time
- Enables remote access to data

Examples:

- Cloud Storage Services (Google Drive, OneDrive): Multiple servers handle data storage and

processing.

- Social Media Platforms: Facebook, Instagram, X (Twitter) processes millions of user posts across global servers.
- Sales Data: A supermarket chain processes transactions from all its branches using a central database.

Parallel Data Processing

In parallel processing, large task is divided into smaller sub-tasks that run simultaneously on multiple

processors to speed up processing time.

- Significantly reduces processing time
- Used in high-performance computing
- Requires specialized hardware

Examples:

- Weather Forecasting: Supercomputers analyze weather patterns simultaneously.
- Scientific Simulations: Used in space research and artificial intelligence training.

Data Processing Methods

Data processing methods refer to the techniques or tools used to process data, whether manually, machines, or fully automated.

There are three main data processing methods: manual, mechanical, or electronic.

Manual Processing

This method involves human handling and processing data without machines. It includes writing, filling,

and calculating manually. Example: A small shopkeeper writing sales records in a notebook.

Advantages

- Low Cost: No need for expensive machines or software.
- Easy to Use: No technical skills or training required.
- Flexibility: Can be used in places without electricity or computers.
- Security: Less risk of hacking or cyber threats.

Disadvantages

- Slow and Time-Consuming: Processing large amounts of data takes a long time.
- Prone to Errors: High chance of mistakes due to human input.
- Difficult to Retrieve Data: Finding old records manually is tedious.
- Limited Scalability: Cannot handle large data efficiently.

Mechanical Data Processing

This method uses simple machines like typewriters, calculators, and cash registers to assist in data

processing. Example: A cashier using a mechanical cash register to total up bills.

Advantages

- Faster than Manual Processing: Machines speed up calculations and data entry.
- Reduces Human Error: More accuracy in calculations.

- Easier Record Keeping: Printed documents last longer than handwritten ones.

Disadvantages

- Limited Automation: Still requires a lot of human effort.
- Maintenance Cost: Machines need servicing or replacement.
- Storage Issues: Physical records take up space.

Electronic Data Processing

This method uses computers and software to process data automatically, making it the most efficient method today. Example: A supermarket using barcode scanner and a computer-based inventory system.

Advantages

- High Speed: Computers process data in milliseconds.
- High Accuracy: Reduces human error significantly.
- Large Data Handling: Can process massive amounts of data.
- Easy Data Retrieval: Stored data can be accessed instantly.
- Automation: Reduces manual workload.
- Cost-Effective in the Long Run: It saves time and money.

Disadvantages

- High Initial Cost: Expensive to set up hardware and software.
- Technical Knowledge Required: Users need training to operate computers.
- Security Risks: Data can be hacked, lost, or corrupted.
- Dependence on Power and Internet: System fail without electricity or network access.

Week 7: Computer Operations

Operating a computer system involves a number of procedures that makes sure the system runs effectively and securely. This entails handling files, correctly turning the system on and off, and carrying out necessary maintenance like formatting storage media. These operations aid in preventing hardware damage, data loss, and system breakdowns.

Some operations that can be carried out on a computer includes:

Starting Up (Booting the Computer)

Booting is the process of turning on a computer and loading the operating system (OS). It can be categorized into two types:

- Cold Booting: Starting a computer that is completely turned off.
- Warm Booting: Restarting a computer that is already on.

Steps to Start a Computer Properly

1. Check Power Supply: Make sure the computer is connected to a power source. If using a laptop, make sure the battery is charged.
2. Turn On the Computer: Press the power button on the system unit or the laptop.
3. POST (Power-On Self-Test): The system checks its hardware components (e.g. RAM, hard drive,

etc) for issues.

4. Load the Operating System: The operating system starts loading, displaying a welcome screen.

5. Login (If Required): Enter a password (if set) to access the password.

Shutting Down the Computer

Turning off the computer properly is important to prevent data loss and system corruption.

Steps to Shut Down the Computer Properly

1. Close All Applications: Save your work and close all open programs.
2. Access the Shutdown Menu: Click the Start Menu, select Power, then click Shut Down.
3. Wait for Shutdown Process: The system will close processes and turn off automatically.
4. Turn Off External Devices: If you're using a desktop, turn off other peripherals like monitor, printer, speaker, and unplug if necessary.

Restarting the Computer

Restarting (or rebooting) a computer is necessary when installing updates, troubleshooting problems, or refreshing the system.

Steps to Restart a Computer

1. Click Start Menu → Power → Restart.
2. The computer will shut down and turn on automatically.

Formatting a Storage Device (Hard Drive, USB, SD Card)

Formatting a storage device prepares the storage device for use by erasing all data and setting up a file

system. It is necessary when installing a new operating system or fixing a corrupted drive.

Steps to Format a USB Drive

1. Insert the USB drive into the computer.
2. Open <This PC> or <My Computer> and locate the drive.
3. Right click on the drive and select Format.
4. Select the File System: NTFS (for Windows), FAT32 (compatible with other OS).
5. Click <Start> and proceed.
6. Wait for the process to complete.

NB: Formatting erases all the data in the storage device, so make sure you back up important files before formatting.

Basic File Operations

Beyond startup and shutdown, users need to understand how to carry out basic file operations.

a. Creating and Saving File

- Open a program (e.g. Microsoft Word).
- Type your content.
- Click File → Save and choose a location (or use Ctrl + S).
- Give the file a name and select a format (e.g. .docx, .pdf).

b. Copying and Deleting Files

- Copy: Right click on a file. Select Copy. Go to destination folder, right click and paste.
- Delete: Right click on a file Select Delete. The file moves to the Recycle Bin.

Installing and Uninstalling Software

a. Installing Software

- Download the setup file (.exe file).
- Double click the file to open the installer.
- Follow the installation instructions.
- Click Finish after installation.

b. Uninstalling Software

- Go to Control Panel → Programs and Features.
- Locate the software.
- Click Uninstall and follow prompts.

Week 8 & 9: Computer Security

Computer security, also called cybersecurity, is the protection of computer systems, networks, and information from harm, theft, unauthorized use, damage, and disruption. It involves implementing security measures to safeguard hardware, software, and information from cyber threats such as viruses, hacking, phishing, and malware attacks.

Computer hardware is typically protected by the same means used to protect other valuable or sensitive

equipment—namely, serial numbers, doors and locks, and alarms.

Objectives of Computer Security

1. Confidentiality: Ensuring that sensitive information is accessible only to authorized users.
2. Integrity: Maintaining the accuracy and reliability of data, preventing unauthorized modification.
3. Authentication: Verifying the identity of users before granting access.
4. Availability: Ensuring that computer systems and data remain accessible when needed.

Key Aspects of Computer Security

1. Access Control: This involves managing who has access to what resources within a system or network. Access control mechanisms include passwords, biometric authentication, access control lists (ACLs), and role-based access control (RBAC).
2. Encryption: Encryption is the process of converting data into a format that cannot be easily read or understood by unauthorized users. It helps protect data confidentiality by scrambling information so that only authorized parties can access it.
3. Firewalls: Firewalls are security devices or software that monitor and control incoming and outgoing network traffic based on predetermined security rules. They serve as a barrier between a trusted internal network and untrusted external networks, such as the internet, filtering the traffic to prevent unauthorized access.
4. Antivirus and Anti-Malware Software: These programs are designed to detect, prevent, and remove malicious software (malware) such as viruses, worms, Trojans, and ransomware from computer systems.
5. Intrusion Detection and Prevention Systems (IDPS): IDPS monitor network or system activities

for malicious activities or policy violations and can take actions to prevent or mitigate potential threats.

6. Patch Management: Regularly updating software and firmware with patches provided by vendors

helps fix known vulnerabilities and protect systems from exploitation by attackers.

7. Security Policies and Procedures: Establishing and enforcing security policies and procedures help

ensure that employees and users follow best practices for maintaining security, such as password management, data handling, and incident response protocols.

8. Incident Response Plan (IRP): A set of procedures to detect, respond to, and recover from security

breaches.

9. Disaster Recovery Plan (DRP): A set of plan to ensure business continuity in case of cyberattacks, hardware failures, or natural disasters.

Basic Safety for Computer Room

The safety of a computer room is the safety of computer systems and other things found in a computer room.

1. A computer room must be spacious.
2. A computer room must be well ventilated.
3. A computer room must be clean and well lit.
4. A computer room temperature must be cool.
5. No one must eat or drink inside the computer room, not even the teacher.
6. Each computer system must have their own UPS.
7. Backup power must be available.
8. Fire extinguisher must be available in a computer room.
9. Safety rules must be visible for all users.
10. Turn off the power and unplug equipment before performing service.

Tips to Minimize Risk of Computer-Related Accident:

- i. Site your computer near an electric socket to avoid trailing wires across the floor; if you use an extension cable, make sure it doesn't overheat and nobody can trip over it.
- ii. Take care not to overload electric sockets; use trailing multi-socket units rather than plug adapters.
- iii. Always follow installation and service instructions in your computer guidebook closely. If in doubt, leave it to the experts.
- iv. Electricity and water do not mix - keep drinks and plants well away from computers.
- v. Regularly check all electrical equipment for damaged plugs or frayed cables.
- vi. Computers are large and bulky pieces of equipment, move them only if you feel confident in doing so, and with care, especially up and down stairs. Use a trolley and a lifter and ask for help.
- vii. Do not allow children to move computers.
- viii. Do not allow children to play on or with computer swivel chairs.

ix. Make sure the computer is sited in a position where you have plenty of room to move and to get out of the room in an emergency.

Malware Infection and Prevention

Malware, short for malicious software, is a broad category of software designed to harm, exploit, or

otherwise compromise the data and systems of its victims. Common types of malware include viruses,

worms, trojans, ransomware, spyware, adware, and rootkits. Each type of malware operates differently

but often has overlapping features and objectives, such as stealing information, disrupting operations, or

extorting money.

Common types of malwares include:

1. Viruses: Malicious code that attaches itself to legitimate programs and replicates when the host program runs.
2. Worms: Self-replicating malware that spreads through networks without user intervention.
3. Trojans: Malicious programs disguised as legitimate software that trick users into executing them.
4. Ransomware: Encrypts files and demands payment for the decryption key.
5. Spyware: Secretly monitors and collects user activity and data.
6. Adware: Automatically displays or downloads advertising material.
7. Rootkits: Designed to gain unauthorized root or administrative access to a system.

Infection Vectors

Infection vectors are the different ways malware or cyber threats enter a system, network, or device.

Hackers and cybercriminals use these methods to gain access, steal data, or cause damage.

Common infection vectors include:

1. Phishing Emails & Malicious Links: An email that looks legitimate but contains a link or attachment that, once clicked, installs malware.
2. Malicious Attachments & File Downloads: Downloading a file from an untrusted source can introduce malware into your system.
3. Drive-By Download: Some websites are rigged to install malware the moment you visit them – no clicking is required.
4. Removable Storage Media (USBs, External Drives, CDs/DVDs): Plugging an infected USB or external drive can spread malware automatically.
5. Unpatched Software & Exploits: Hackers love outdated software because security holes make it easy to sneak in and attack the system.
6. Malicious Ads (Malvertising): Even legit websites can be compromised by third-party partners that displays ads that can silently install malware without your knowledge.
7. Social Engineering & Human Error: Sometimes, people unknowingly help hackers by sharing sensitive information or following misleading instructions.

Impact of malware include:

- Data Loss: Critical files can be corrupted, encrypted, or deleted.
- Financial Loss: Ransom payments, fraudulent transactions, or repair costs.
- Operational Disruption: Systems and networks can be rendered inoperable.
- Reputational Damage: Loss of customer trust and business credibility.
- Legal Consequences: Failure to protect data can result in regulatory fines and lawsuits.

Some prevention strategies you can employ to protect your computer and data include:

1. Use Antivirus and Anti-malware Software: Regularly update and scan your system with reputable security software.
2. Regular Updates and Patches: Keep your operating system, software, and firmware up to date to close vulnerabilities.
3. Firewalls: Implement hardware and software firewalls to block unauthorized access.
4. Educate Users: Train users to recognize phishing attacks, suspicious links, and unsafe practices.
5. Email Filtering: Use email filters to block malicious emails and attachments.
6. Backup Data: Regularly back up critical data to restore systems in case of an attack.
7. Use Strong, Unique Passwords: Implement complex passwords and change them regularly.
8. Least Privilege Principle: Grant users the minimum level of access necessary to perform their duties.
9. Network Segmentation: Divide your network into segments to limit the spread of malware.
10. Monitor and Audit: Continuously monitor systems for unusual activity and regularly audit security measures.

Computer Ergonomics

First, what is ergonomics?

Ergonomics is the science of designing work environment, tools, and tasks to fit human capabilities and

limitations. It focuses on improving efficiency, comfort, and safety to reduce strain and prevent injuries.

Computer ergonomics is a branch of ergonomics that deals with creating a comfortable and efficient

workspace for computer users. It aims to prevent strain, discomfort, and injuries caused by prolonged

computer use, such as:

- Repetitive strain injuries (RSI) (e.g. carpal tunnel syndrome)
- Eye strain and fatigue
- Neck, back, and shoulder pain
- Poor posture-related issues

Key aspects of computer ergonomics include:

1. Proper Seating & Posture: Use a chair with a good lumbar support to maintain the natural curve

of the spine. The chair height should be adjustable so that the feet are flat on the floor, and knees

are at 90-degree angle. Sit with back straight and shoulders relaxed to avoid hunching.

2. Monitor Placement: The top of the screen should be at eye level to prevent neck strain. Adjust the screen brightness and contrast to reduce eye strain.

3. Keyboard & Mouse Positioning: Keep the keyboard at elbow level to maintain a natural wrist position. Place the mouse close to the keyboard to minimize arm strain.

4. Lighting & Screen Adjustments: Reduce glare by positioning the monitor away from direct light sources. Use blue light filters or night mode to reduce eye strain. Follow the 20-20-20 rule (every 20 minutes, look at something 20 feet away for 20 seconds).

5. Break & Movement: Take frequent breaks to stand, stretch, and move around. Avoid prolonged static postures by adjusting positions regularly.

Computer ergonomics is important, because:

- It prevents long-term health issues
- It improves productivity and focus by reducing discomfort
- Reduces workplace injuries related to prolonged computer use.
- Enhances overall well-being by promoting better posture and movement.

Week 10 & 11: Computer Network

What is a Computer Network?

A computer network is a group of interconnected devices (such as computers, servers, routers, switches,

and other hardware) that can communicate and share resources (such as data, applications, and hardware

components) with each other. These devices are linked using communication channels such as cables,

wireless signals, or fibre optics. These networks can be set in various configurations and sizes, ranging

from small Local Area Networks (LANs) to extensive Wide Area Networks (WANs) that spans large geographical areas.

Importance of Computer Networks

Computer networks play a crucial role in modern computing. Here are some of its importance:

- Resource Sharing: Enables sharing of printers, files, internet connections, and other resources.
- Communication: Enables communication through emails, video conferencing, instant messaging, and VoIP (Voice over Internet Protocol).
- Data Sharing and Accessibility: Allows users to access and share data easily, regardless of location.
- Cost Efficiency: Reduces hardware and software cost by enabling centralized storage and processing.
- Scalability: Networks can be easily scaled by adding new devices without significant changes to the existing infrastructure.
- Improved Security and Backup: Provides better security mechanisms, access control, and

automated backups for data protection.

- Automation and Control: Networks enable automation of tasks and centralized control of connected devices, improving productivity and operational efficiency.

Types of Network Topologies

Network topology refers to the arrangement of devices and how they are connected in a network. The

common types include:

- Bus Topology:** In this topology, all devices are connected to a single central cable (bus). Devices are easy to install and require less cable. The disadvantage of using this topology is, a single cable failure can bring down the entire network.
- Star Topology:** In this topology, all devices are connected to a central hub or switch. Devices are easy to manage, scale, and failure of one device does not affect other devices. The disadvantage of using this topology is that if the central hub fails, the entire network goes down.
- Ring Topology:** In this topology, devices are connected in a circular loop. The data flow in one direction, reducing the risk of data collision. The disadvantage of this topology is that a single point of failure can break the network.
- Mesh Topology:** In this topology, each device is directly connected to every other device. It is highly reliable with multiple paths for data to flow. The disadvantage of this topology is that it is expensive and complex to set up.
- Tree Topology:** This topology forms a structure where devices are connected hierarchically. It resembles a tree with a root node and various branches. The root node is connected to multiple levels of child nodes, forming a hierarchy. The disadvantage of this topology is that if a node fails, the branches will go down too.
- Hybrid Topology:** This topology is a combination of two or more different topologies. It is flexible and scalable based on network requirements. The disadvantage of this topology is that it can be complex and expensive to maintain.

Types of Networks

Networks can be categorized based on their geographical coverage and purpose:

Personal Area Network (PAN)

A Personal Area Network (PAN) is a small network that connects devices within an individual's workspace,

usually within a range of a few meters. It is used for communication between personal devices such as

smartphones, laptops, tablets, and smartwatches.

Characteristics:

- **Short-range Connectivity:** Covers a small area, usually within 10 meters. Primarily connects to an individual's personal devices.
- **Low Power Consumption:** Designed for personal devices with limited power.
- **Wireless or Wired:** Can use Bluetooth, infrared, USB cables, or Wi-Fi
- **Limited Security Risks:** Since it operates in a small area, it has minimal external threats.

Uses:

- Connecting personal devices like smartphones, laptops, and smartwatches.
- File sharing between nearby devices.
- Wireless connection, such as Bluetooth keyboards, mice, and printers.

Local Area Network (LAN)

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area, such as a home, office, or a small group of buildings (e.g., a school or a university campus).

Characteristics:

- Geographic Scope: Limited to a single building or a small group of buildings.
- Speed: Typically, high speed, ranging from 100 Mbps to 10 Gbps.
- Topology: Common topologies include star, ring, and bus.
- Ownership: Usually owned, managed, and maintained by a single organization or individual.
- Media: Often uses Ethernet cables (e.g., Cat 5e, Cat 6) or wireless technologies (e.g., Wi-Fi).
- Cost: Relatively low setup and maintenance costs.

Uses:

- Connecting personal computers, printers, and other devices within an office or home.
- Enabling file sharing, internet access, and peripheral sharing (e.g., printers).
- Supporting internal communications such as email and messaging systems.

Metropolitan Area Network (MAN)

A Metropolitan Area Network (MAN) is a network that covers a larger geographic area than a LAN but smaller than a WAN, typically spanning a city or a large campus.

Characteristics:

- Geographic Scope: Spans a city or a large campus area.
- Speed: High-speed connections, often in the range of 1 Gbps to 100 Gbps.
- Topology: Can include various topologies such as ring, star, or mesh.
- Ownership: May be owned and operated by a single organization or a consortium of users; sometimes leased from service providers.
- Media: Uses fiber optic cables, microwave links, and sometimes wireless technologies.
- Cost: Higher cost compared to LAN due to larger coverage area and infrastructure.

Uses

- Connecting multiple LANs within a city to form a cohesive network.
- Supporting municipal services like emergency response systems, traffic control systems, and public internet access.
- Facilitating communication and data exchange between various branches of an organization within a city.

Wide Area Network

A Wide Area Network (WAN) is a network that spans a large geographic area, often a country or continent, connecting multiple smaller networks, including LANs and MANs.

Characteristics:

- **Geographic Scope:** Can cover large regions, countries, or even continents.
- **Speed:** Variable speeds depending on the technology and infrastructure, ranging from 56 Kbps (dial-up) to several Gbps (fiber optics).
- **Topology:** Often uses mesh, star, or hybrid topologies.
- **Ownership:** Typically owned and operated by multiple organizations or service providers; often involves leasing services from telecom companies.
- **Media:** Utilizes a variety of transmission media, including fiber optic cables, satellite links, and undersea cables.
- **Cost:** High setup and operational costs due to extensive infrastructure and long-distance data transmission.

Uses:

- Connecting geographically dispersed offices and branches of an organization.
- Facilitating global communication and data transfer.
- Supporting the infrastructure of the Internet, which is the largest WAN.
- Enabling remote access to organizational resources and services.

Virtual Private Network (VPN)

A Virtual Private Network (VPN) is a secure network connection that enables users to access the internet

or private networks safely by encrypting data and masking their IP addresses. It is commonly used to

enhance security, privacy, and remote access to corporate networks.

Characteristics:

- **Encryption:** Data is securely encrypted to protect against cyber threats.
- **Remote Access:** Allows users to securely access private networks from anywhere.
- **Anonymity:** Hides the user's real IP address, improving privacy.
- **Bypasses Geo-Restrictions:** Enables access to restricted content in different regions.

Uses:

- Secure remote work by enabling employees to access company networks safely.
- Protecting online privacy by masking IP addresses and encrypting data.
- Accessing restricted content by bypassing regional internet restrictions.
- Preventing cyber threats when using public Wi-Fi networks.

Various LAN Components

1. **Network Interface Cards (NICs):** A network interface card (NIC) is a hardware component that is used to connect computers and devices to a network and allow them to communicate. It can be built into the motherboard or installed as an expansion card. NICs can be wired (Ethernet) or Wireless (Wi-Fi).

2. Switch: A switch is a device that receives, processes, and forwards data to the destination device on the same network. It operates at the data link layer (Layer 2) of the OSI model. It uses MAC (Media Access Control) addresses to forward data efficiently.
3. Hub (Legacy Component): A hub is an outdated networking device that broadcast data to all connected devices. Unlike switches, hubs do not filter traffic, leading to network congestion.
4. Router: A router is a device that directs data packets between networks, such as between a LAN and the Internet. It operates at the network layer (Layer 3) of the OSI model. Routers can be Wired routers and wireless routers, with capabilities like Firewall, Network Address Translation (NAT), and Dynamic Host Configuration Protocol (DHCP).
5. Access Point (AP): An access point is a device that allows wireless devices to connect to a wired network using Wi-Fi. It connects to a switch or router to extend the network wirelessly. It acts as a bridge between the wired and wireless segments of the network.
6. Cables and Connectors: Various types of cables are used in LANs to connect devices, such as Ethernet cables for wired connections and fiber optic cables for high-speed connections. RJ-45 connectors are used to terminate Ethernet cables.
7. Servers: A server is a device that provides network services to other computers (clients) on the network, such as file storage, printing, or authentication. Common types of servers include file servers, print servers, database servers.
8. Clients: Clients are the computers and devices that users interact with. They connect to the network to use shared resources and services. Examples are Desktop computers, laptops, smartphones, and tablets.
9. Network Operating System (NOS): A network operating system is specialized software that manages network resources, users, and devices. It provides services like file sharing, printer access, user management, and security. Examples include Microsoft Windows Server, Linux-based systems, and Novell NetWare.
10. Firewall: A firewall is a security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules. It protects the LAN by filtering incoming and outgoing traffic. It can be hardware-based (part of a router) or software-based.
11. Modem: A modem is used to connect a LAN to an Internet Service Provider (ISP) like MTN, Glo, for internet access.